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(54) **DISCONTINUOUS RECEPTION WITH
IMPLICIT INDICATION IN SIDELINK**

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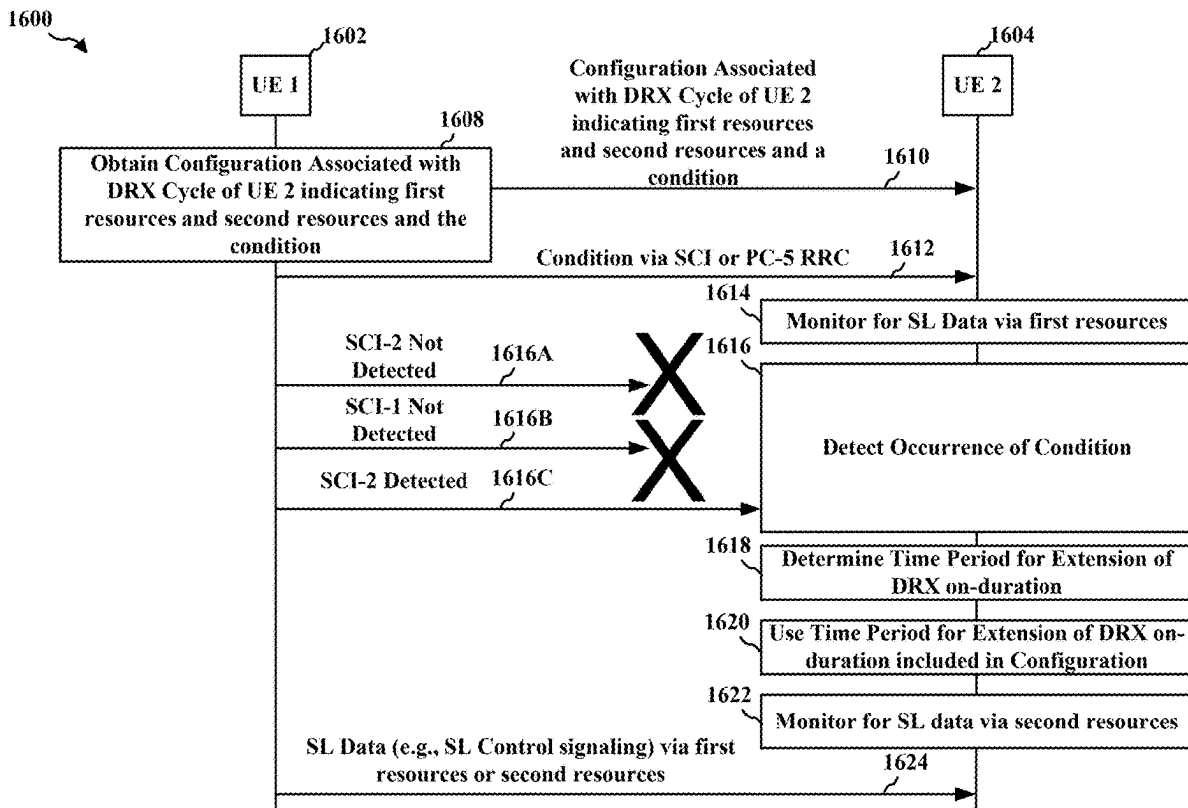
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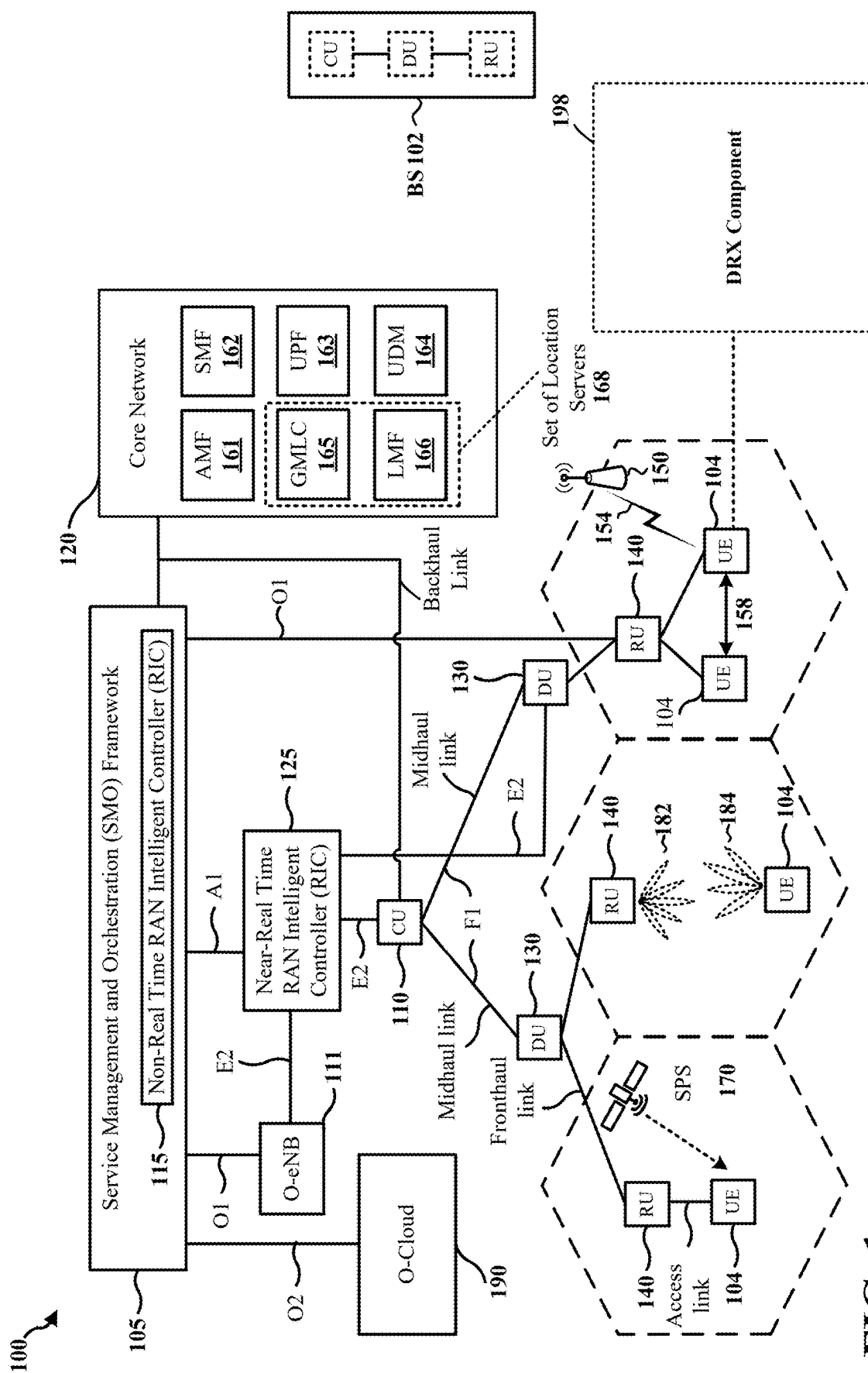
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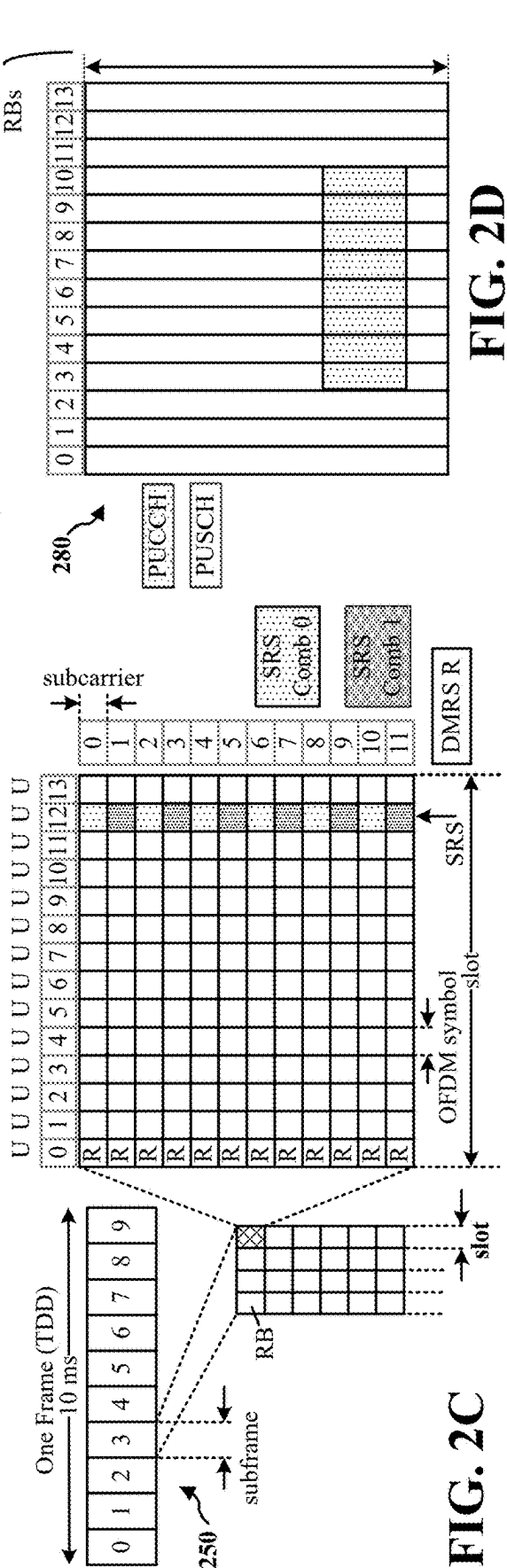
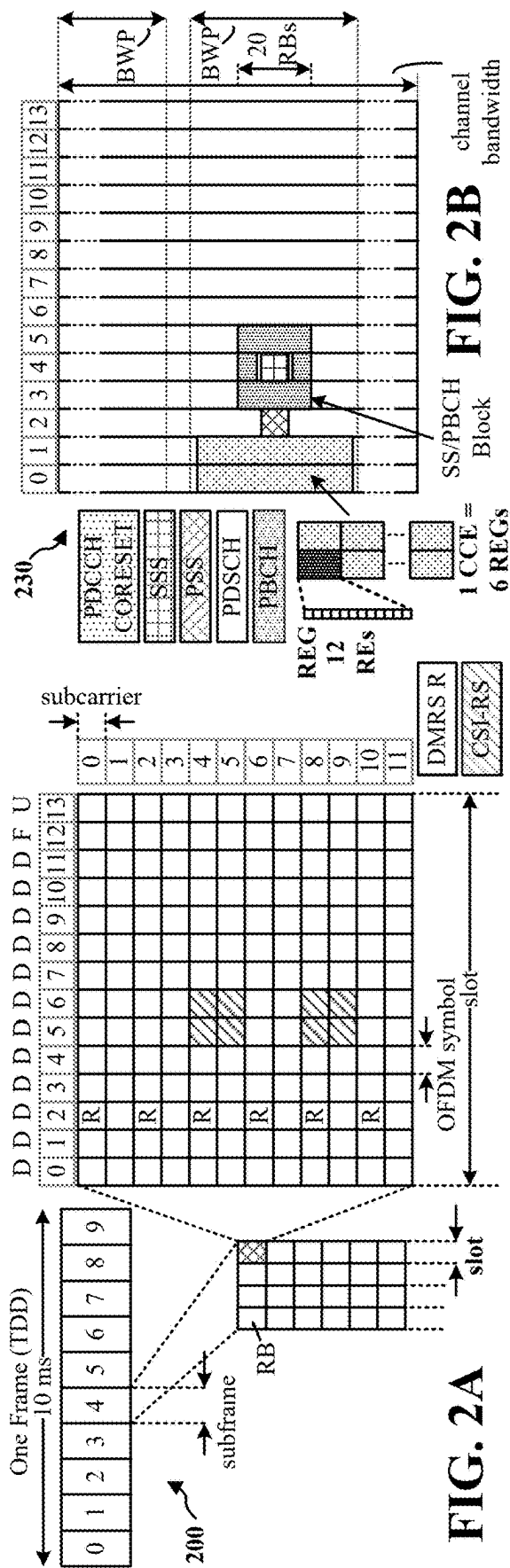
(2) Date: **Feb. 19, 2025**

(57) **ABSTRACT**

A method of wireless communication at a first UE is disclosed herein. The method includes obtaining a configuration associated with a DRX cycle of a second UE, where the configuration includes a first indication of a first set of resources and at least one condition, where the first set of resources is different from a second set of resources utilized for the DRX cycle of the second UE. The method further includes transmitting, to the second UE based on an occurrence of the at least one condition, SL data via the first set of resources or the second set of resources.







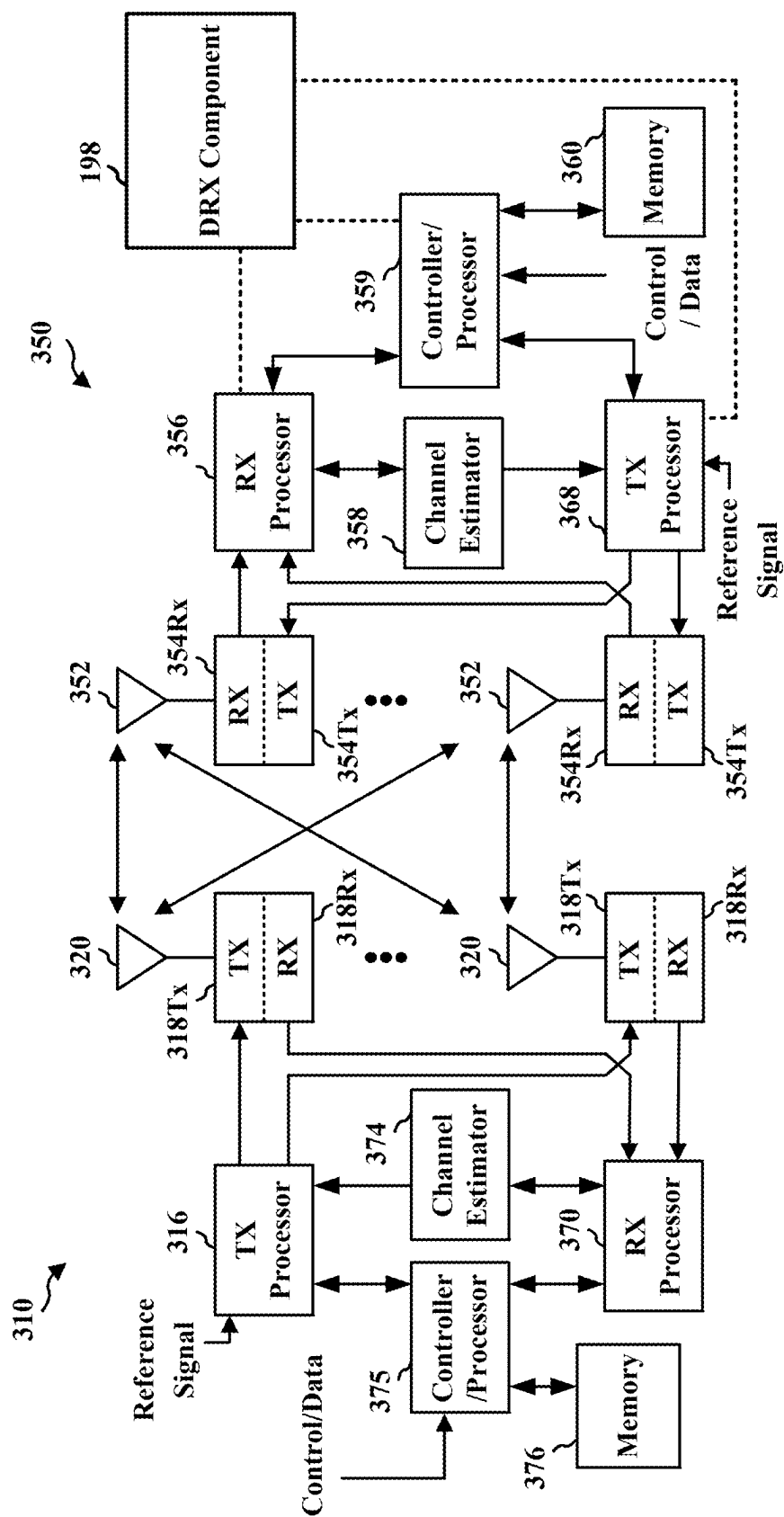


FIG. 3

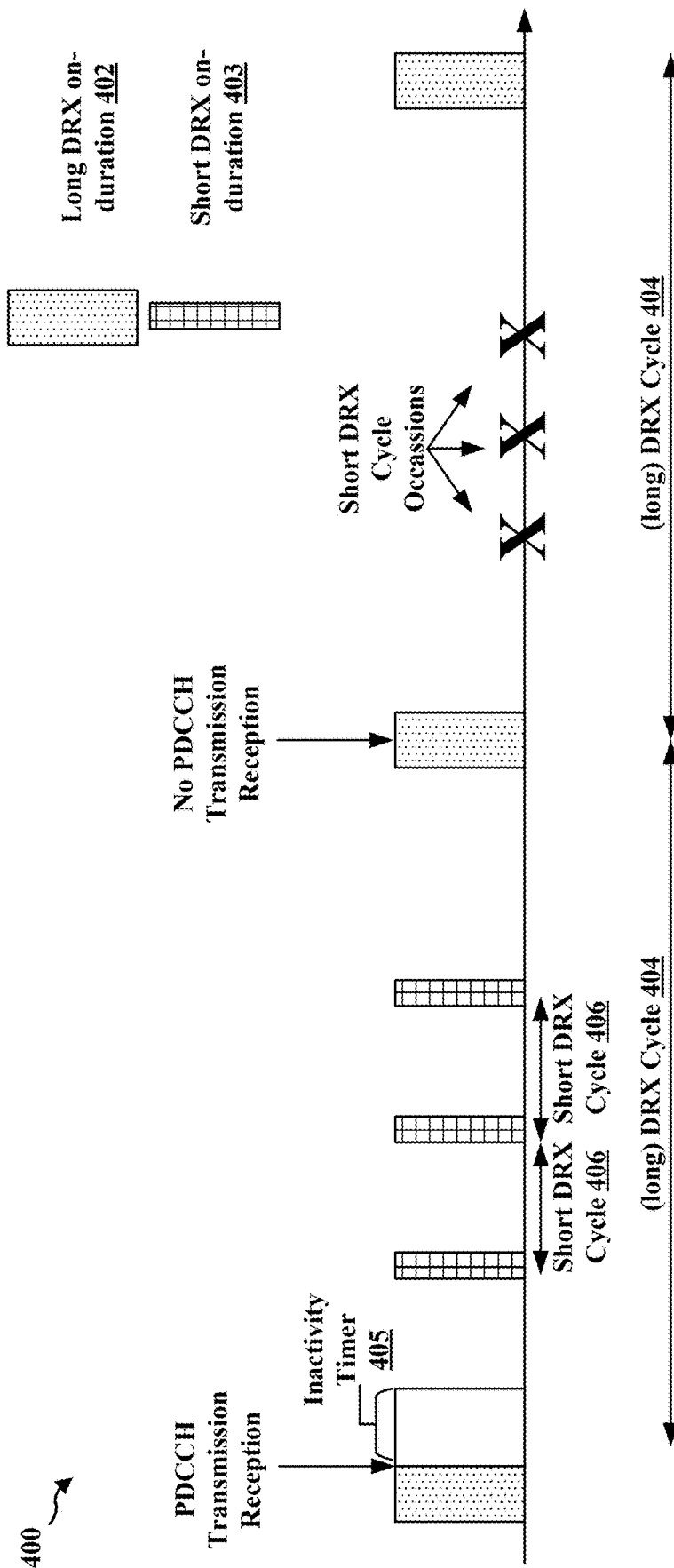


FIG. 4

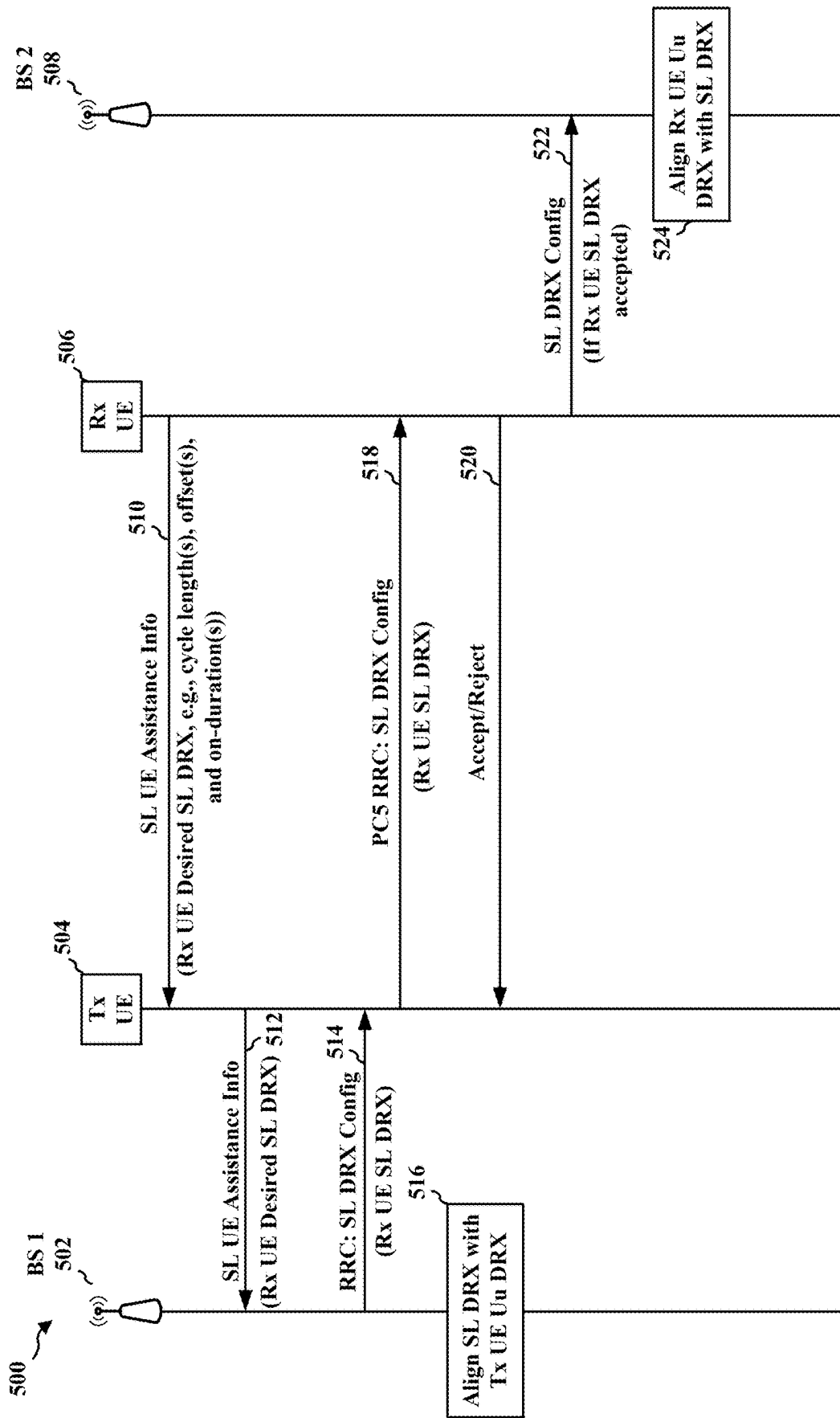


FIG. 5

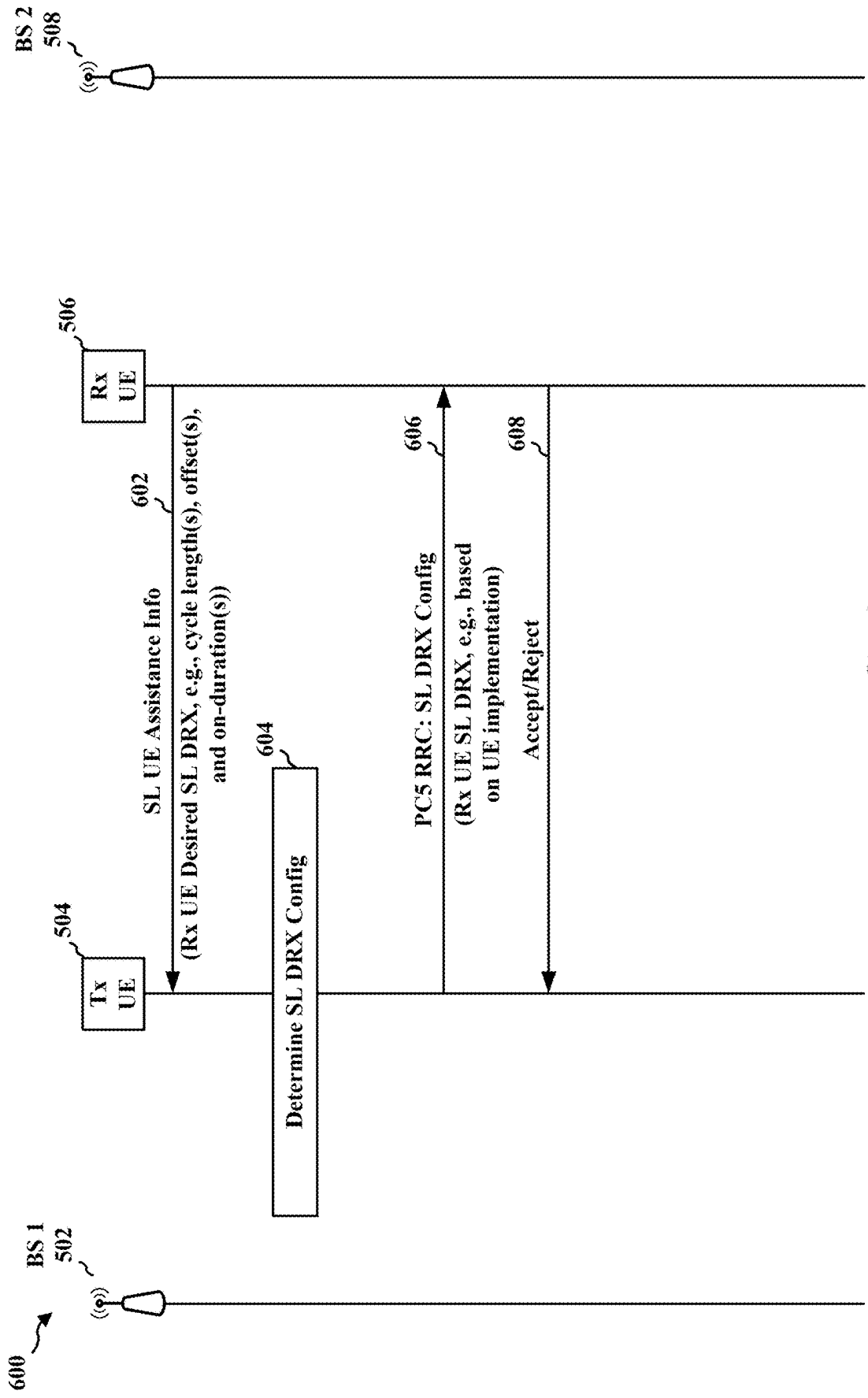


FIG. 6

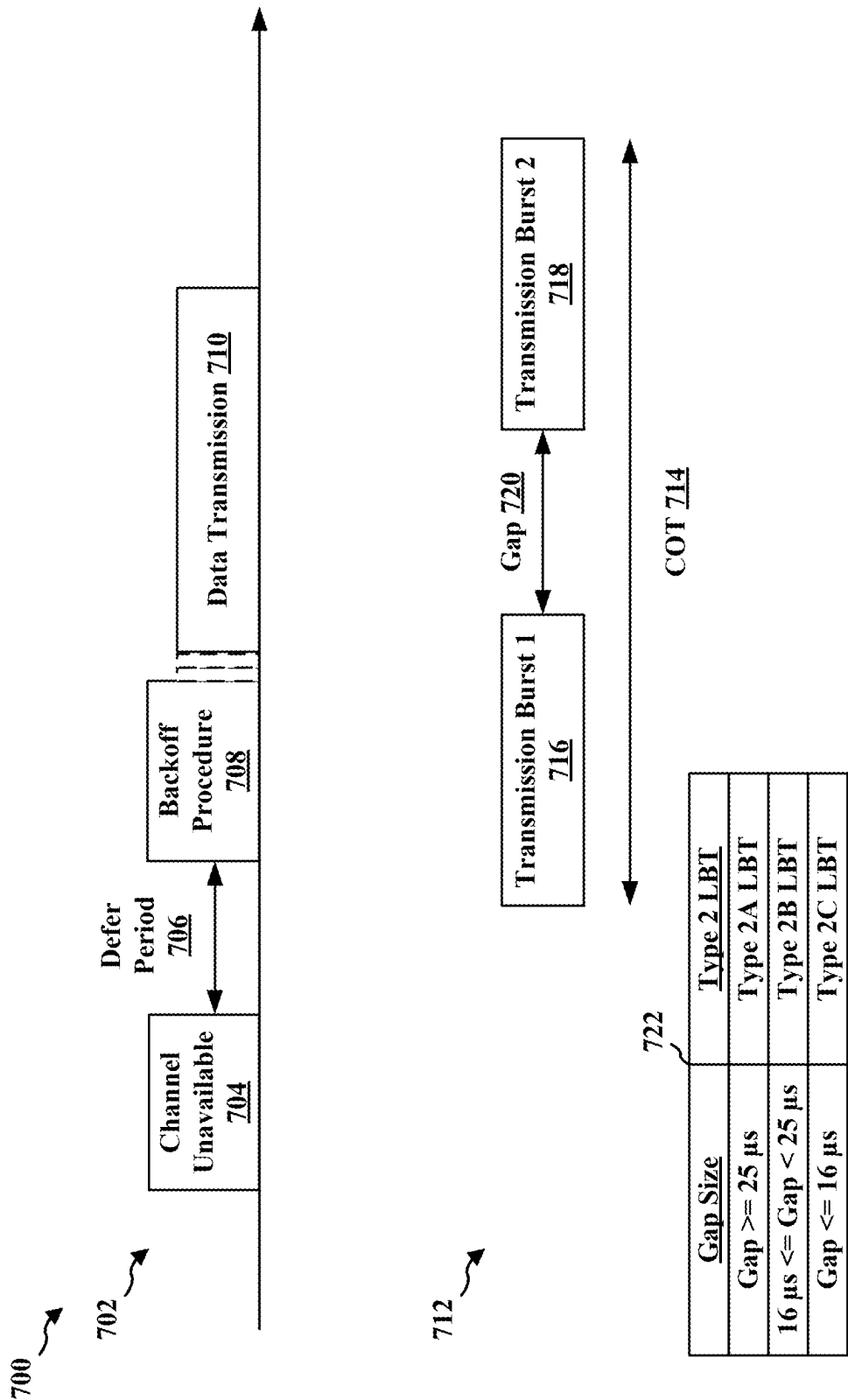


FIG. 7

800 ↗

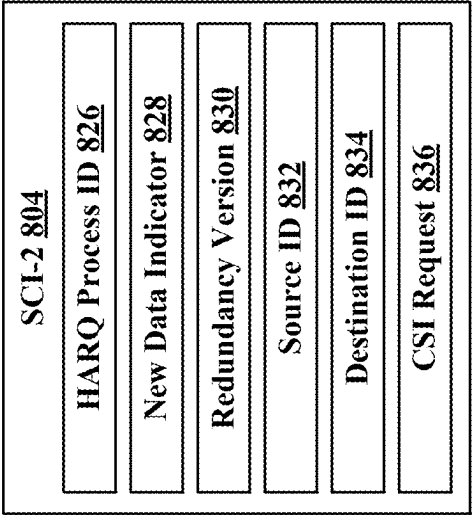
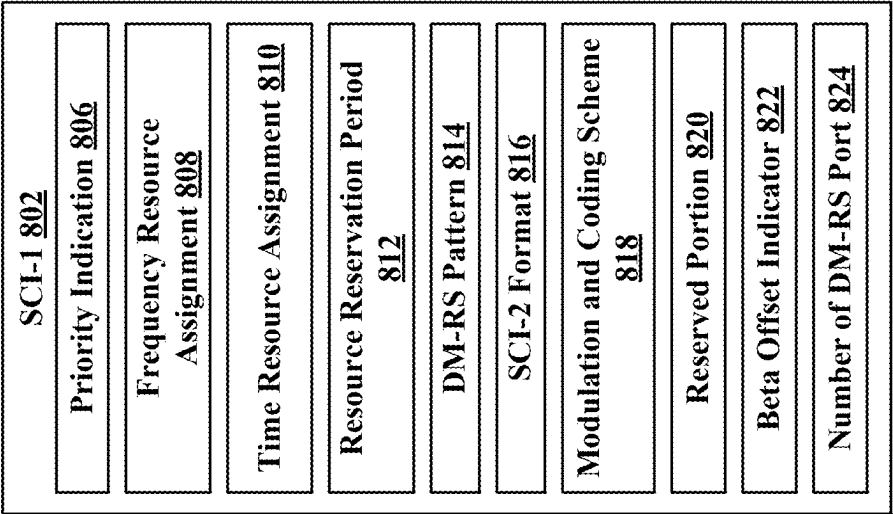


FIG. 8

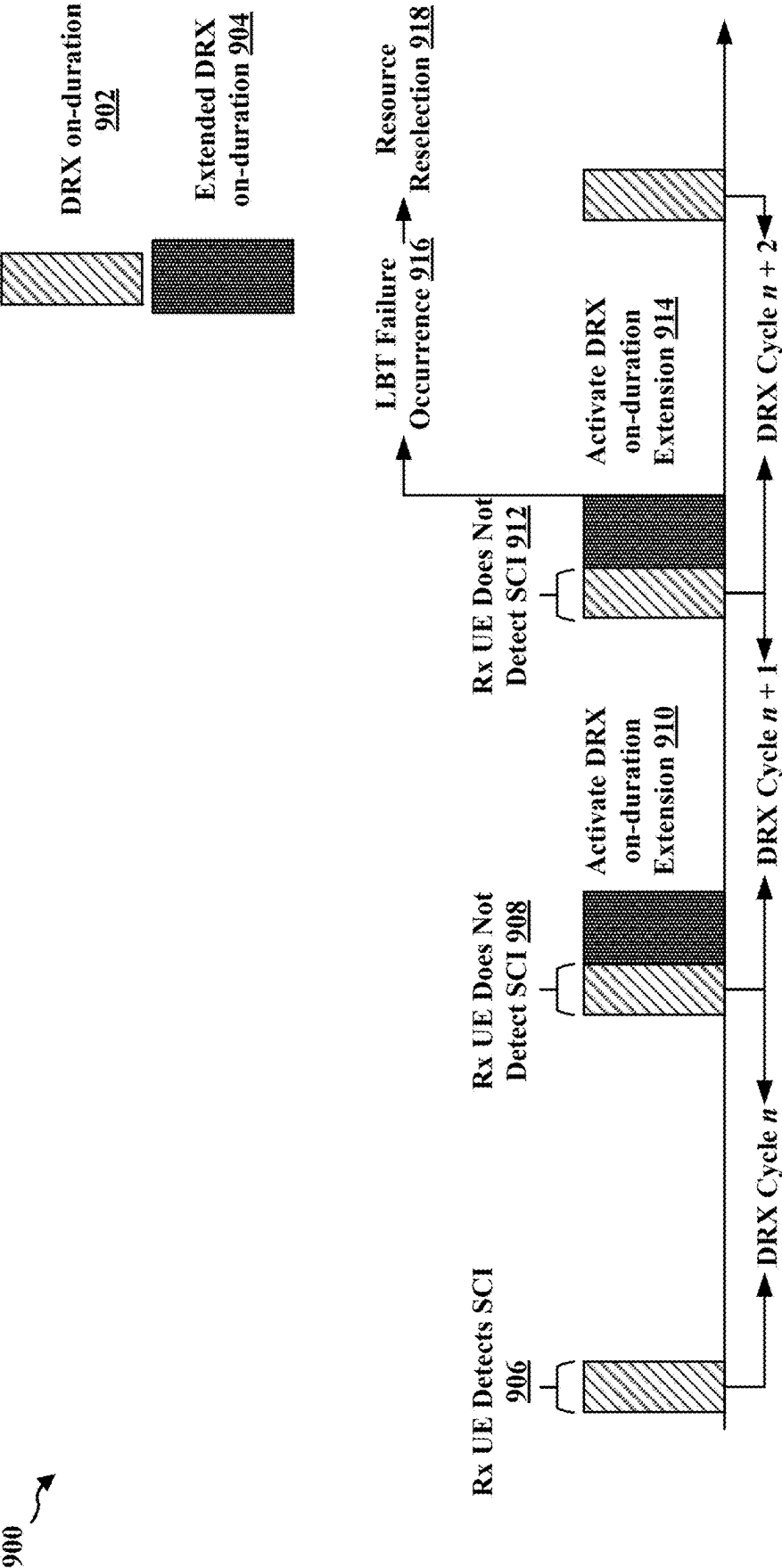


FIG. 9

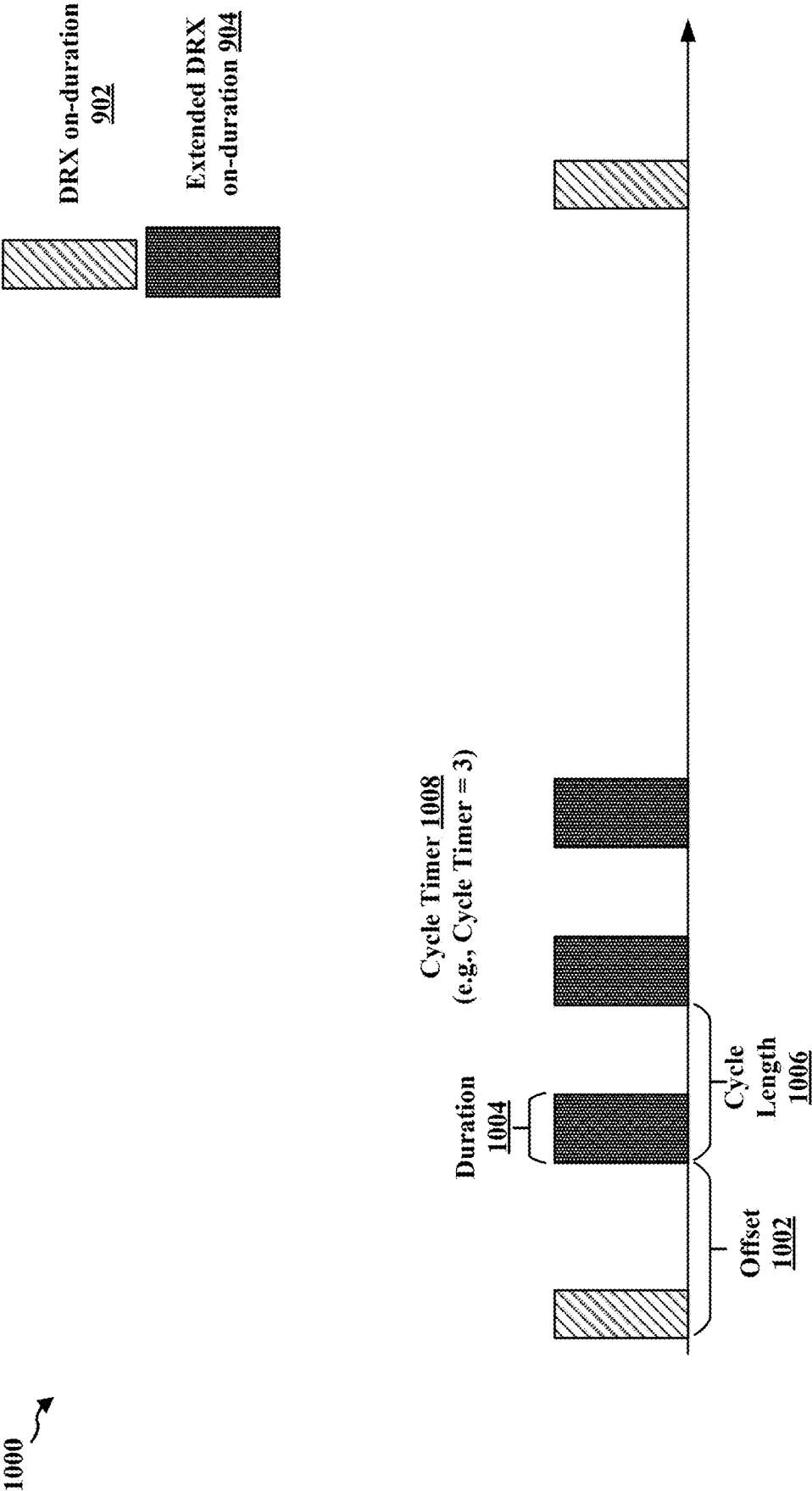


FIG. 10

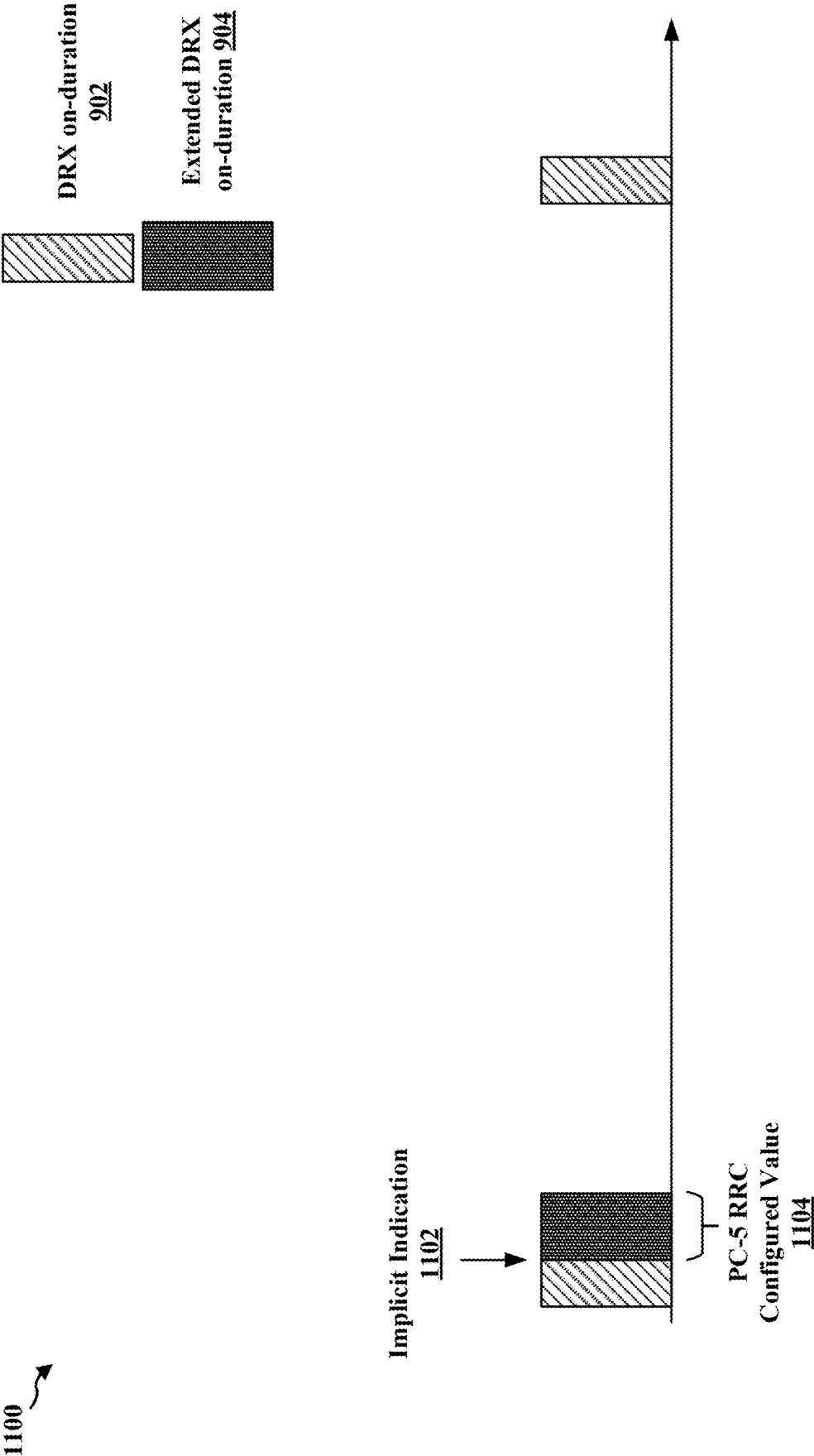


FIG. 11

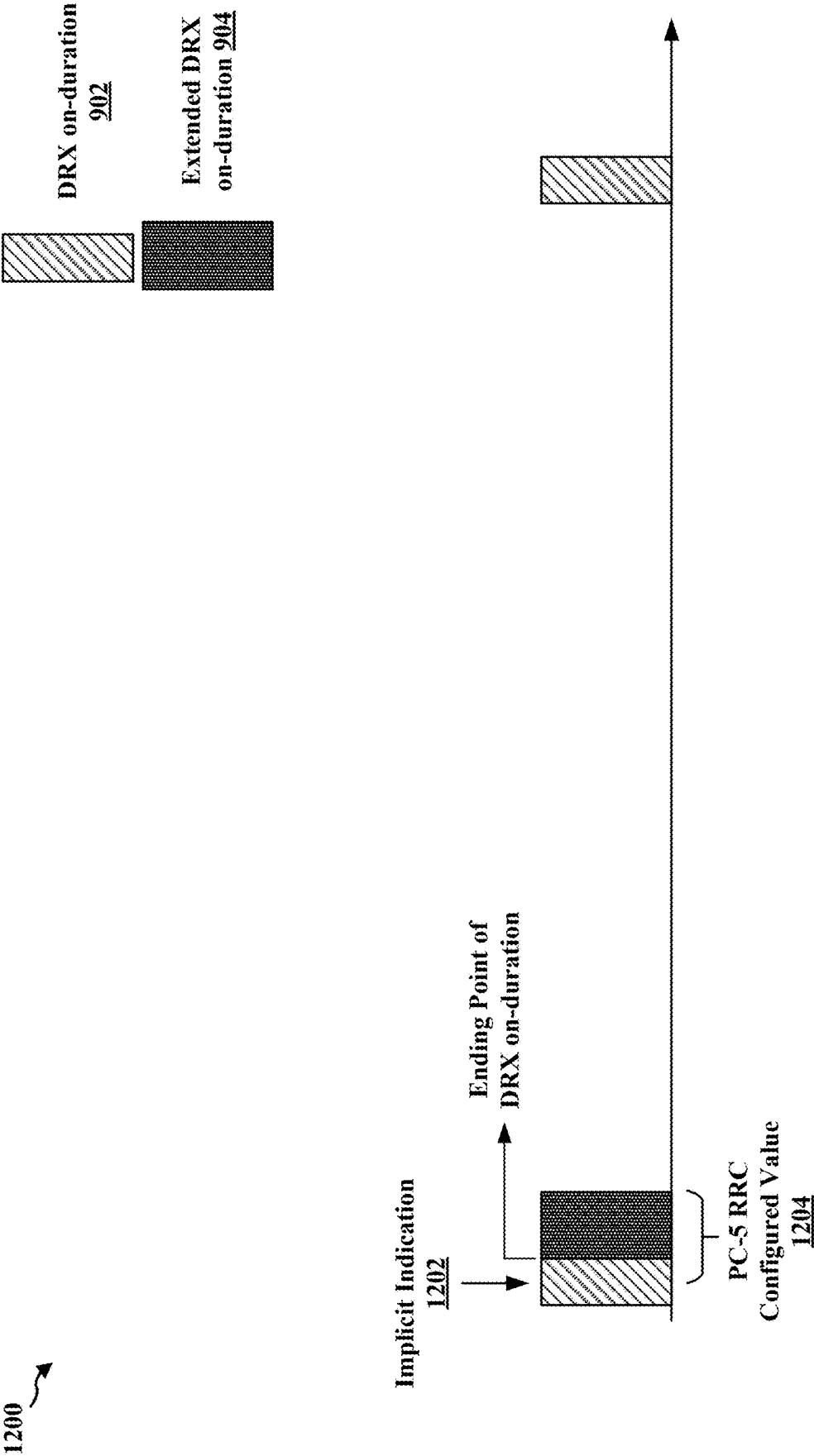


FIG. 12

1300 ↗

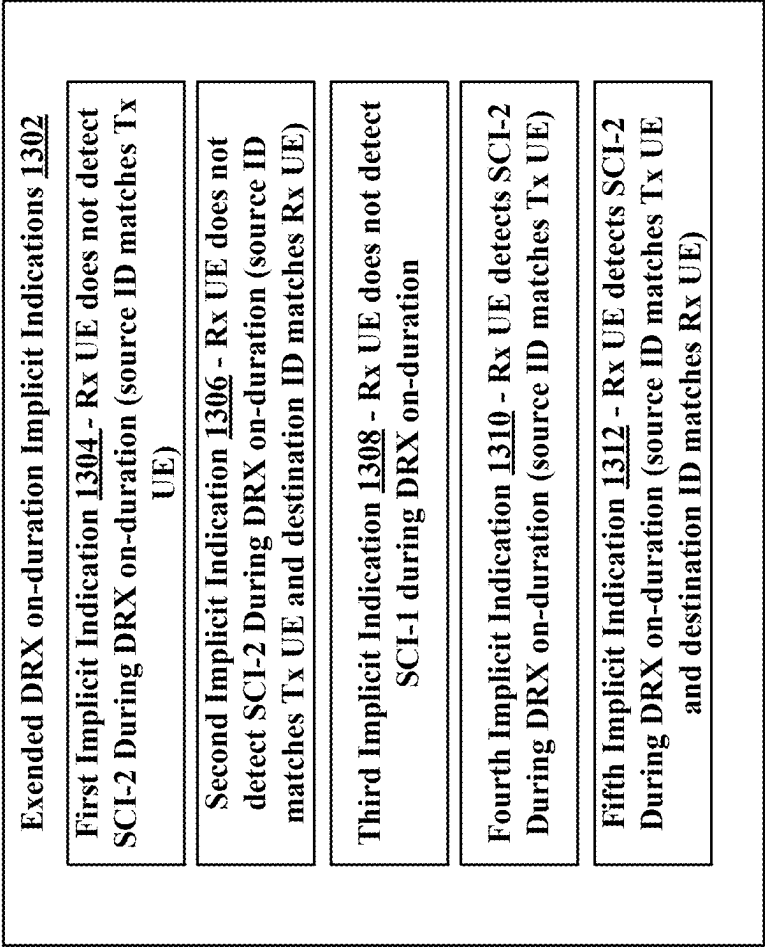


FIG. 13

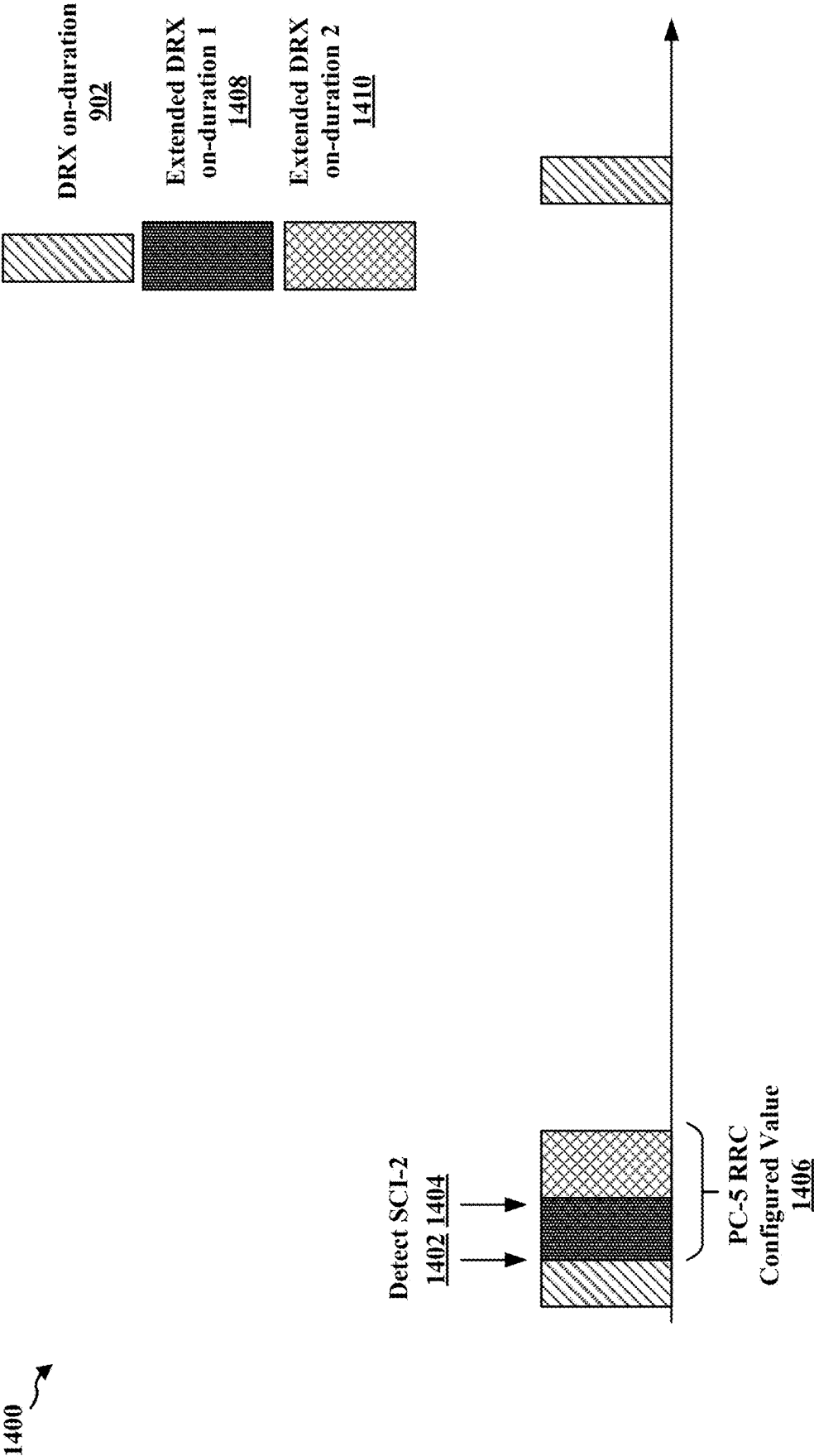


FIG. 14

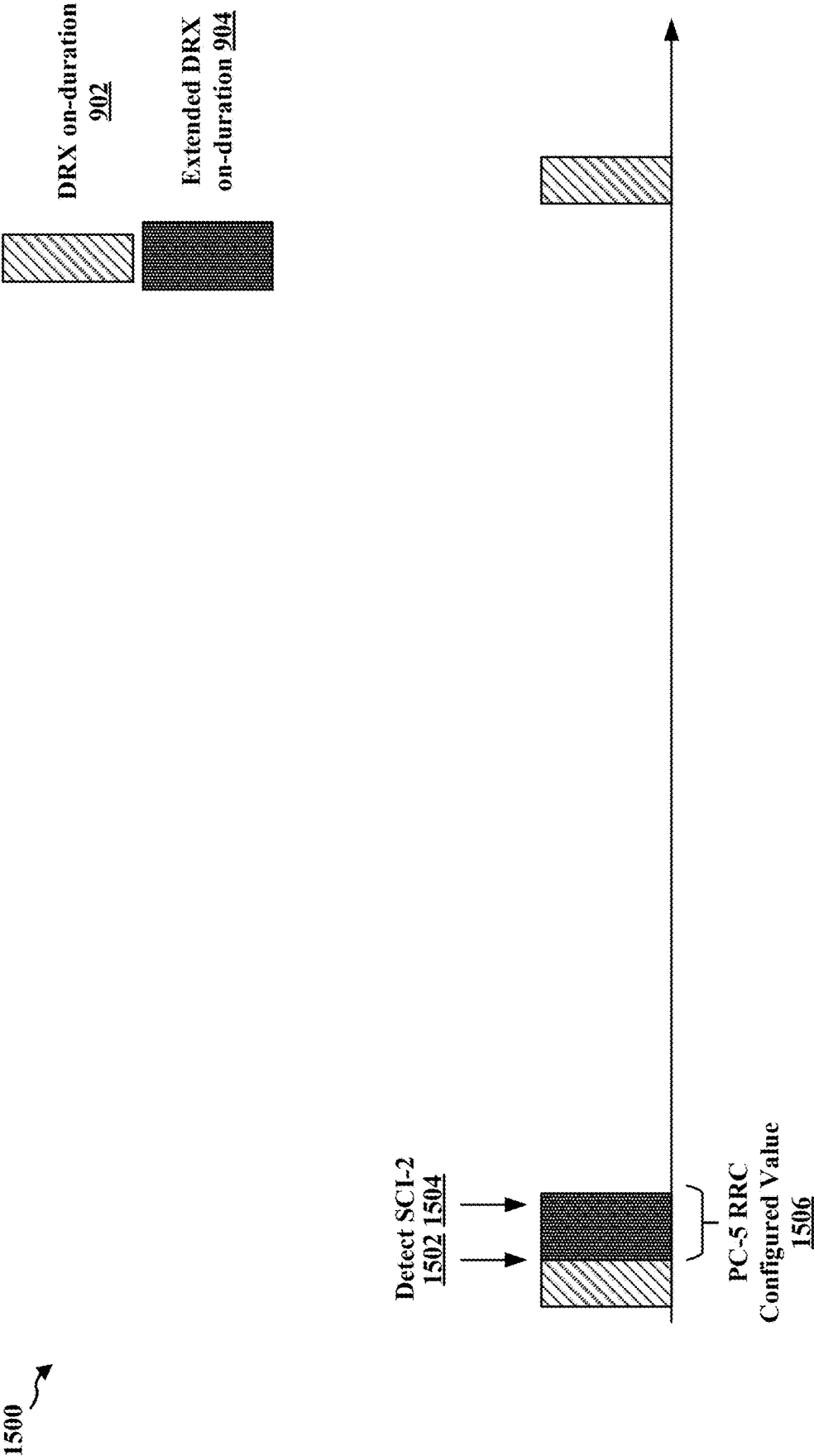


FIG. 15

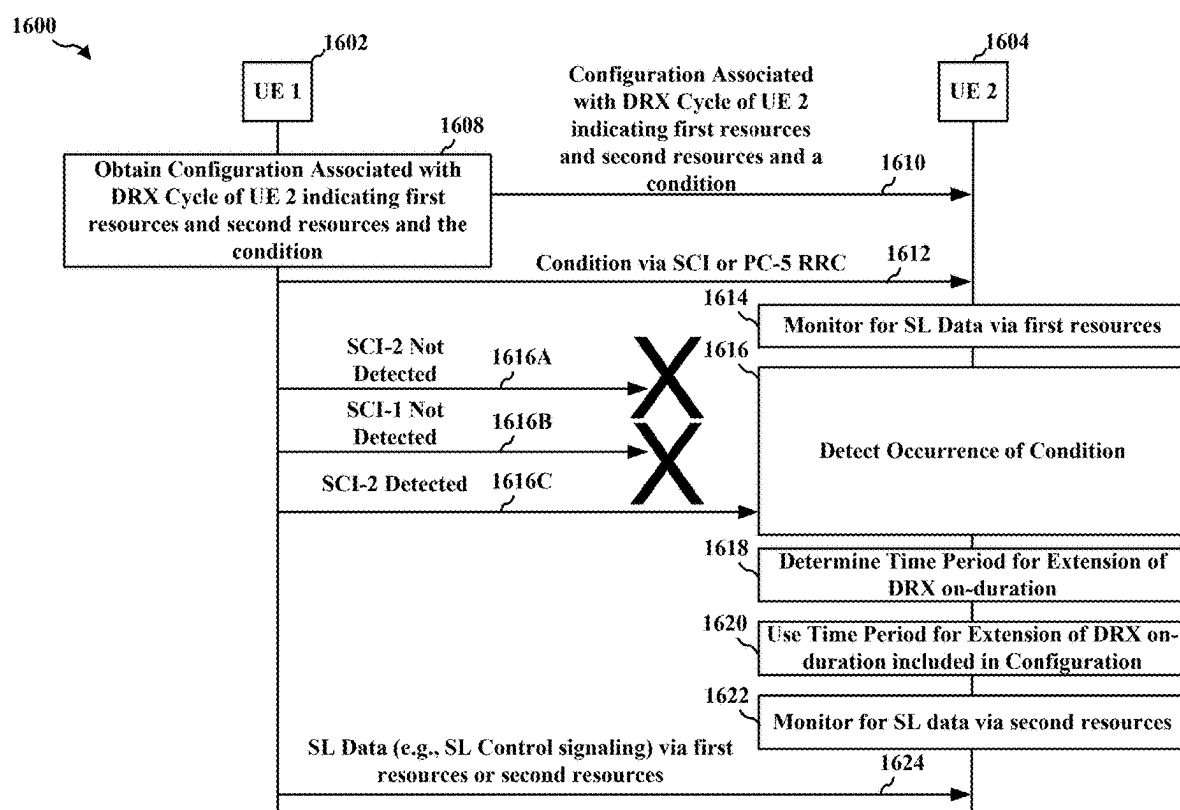
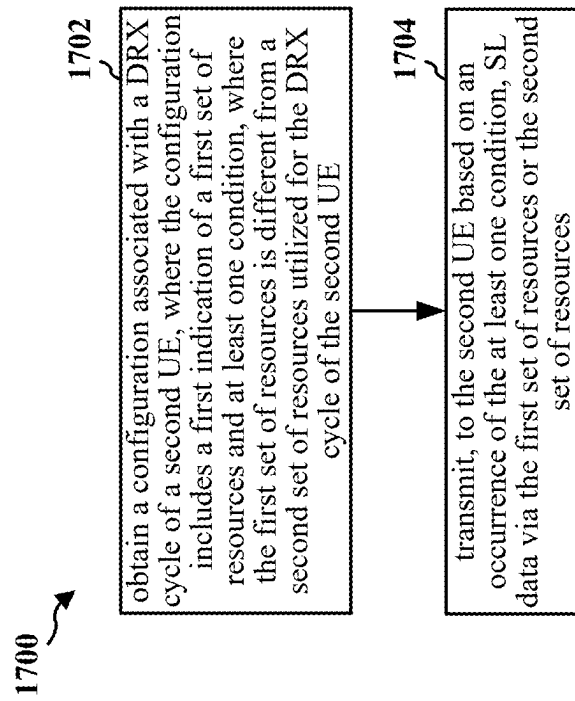
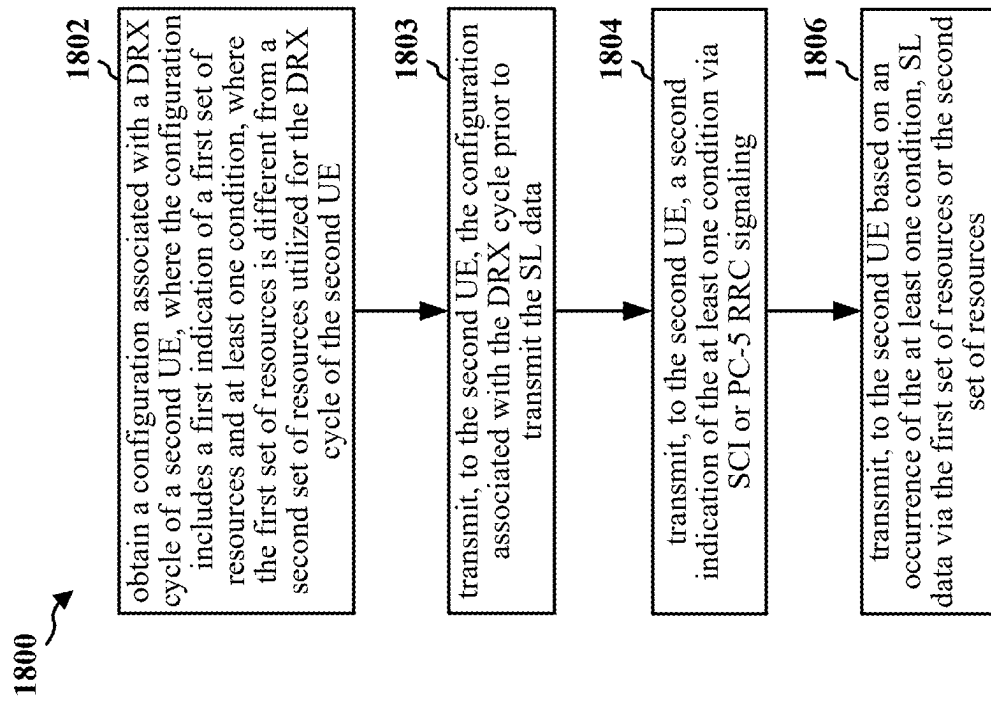


FIG. 16

**FIG. 17**

**FIG. 18**

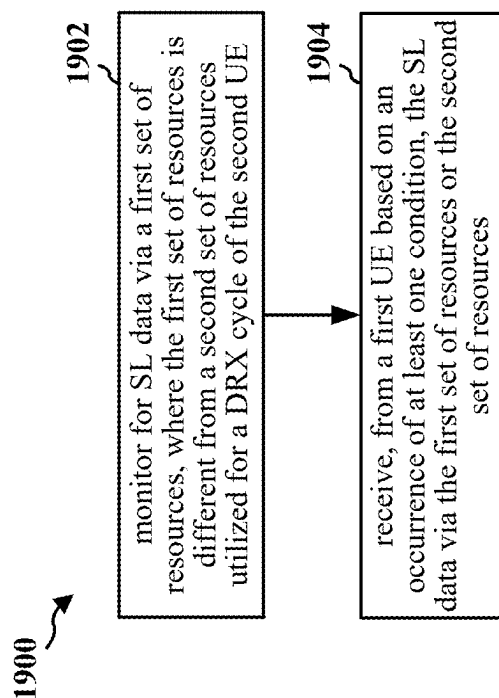


FIG. 19

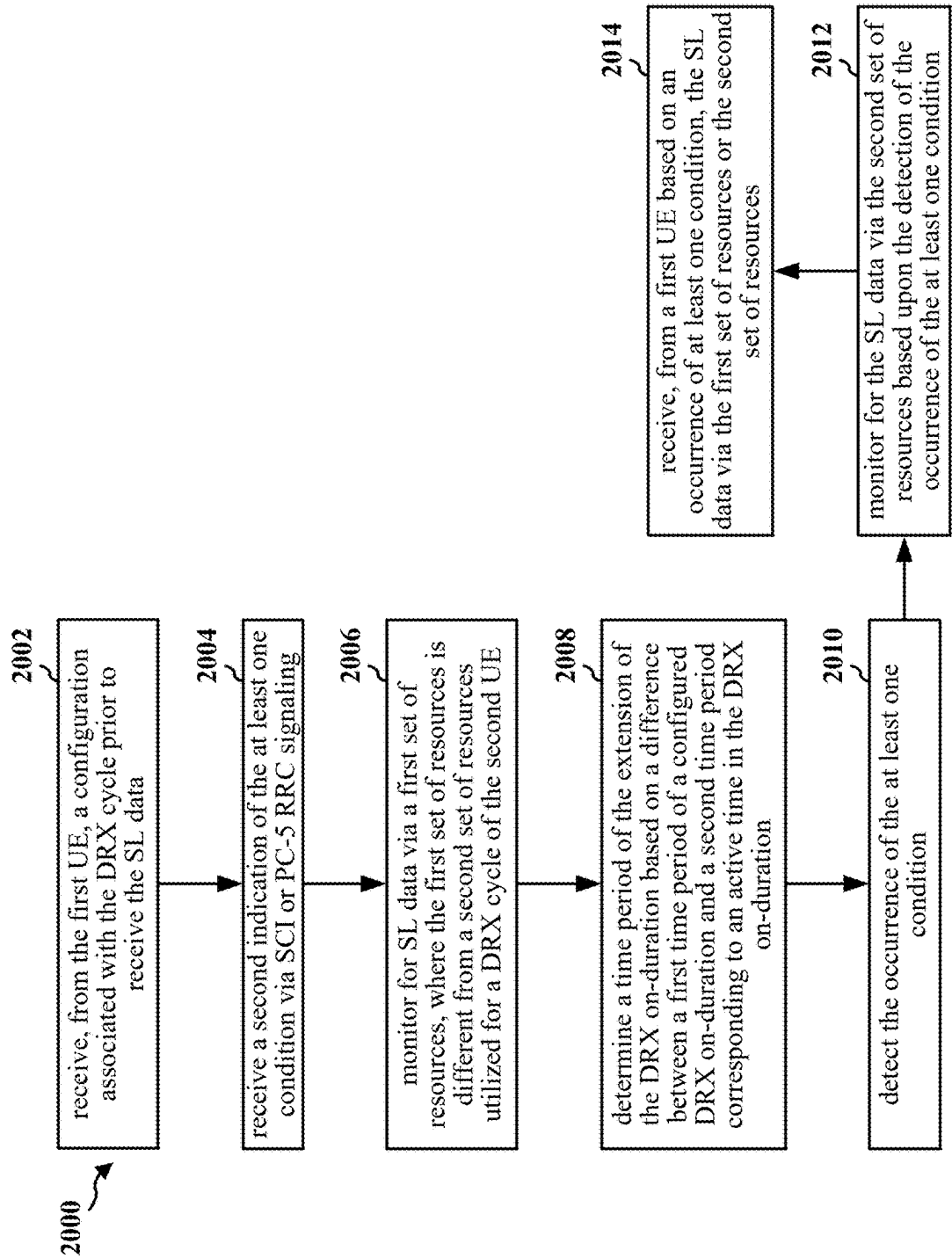
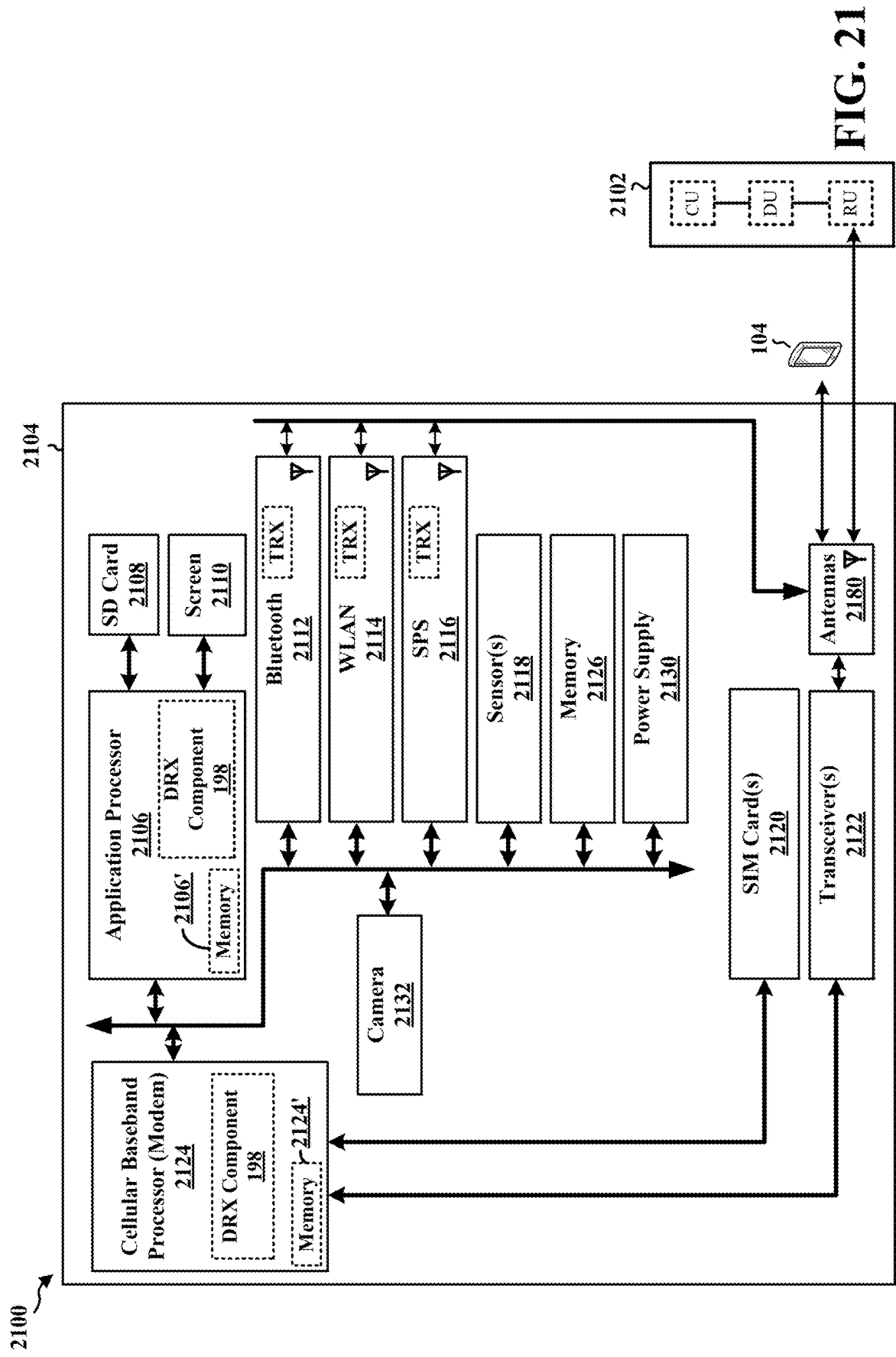


FIG. 20



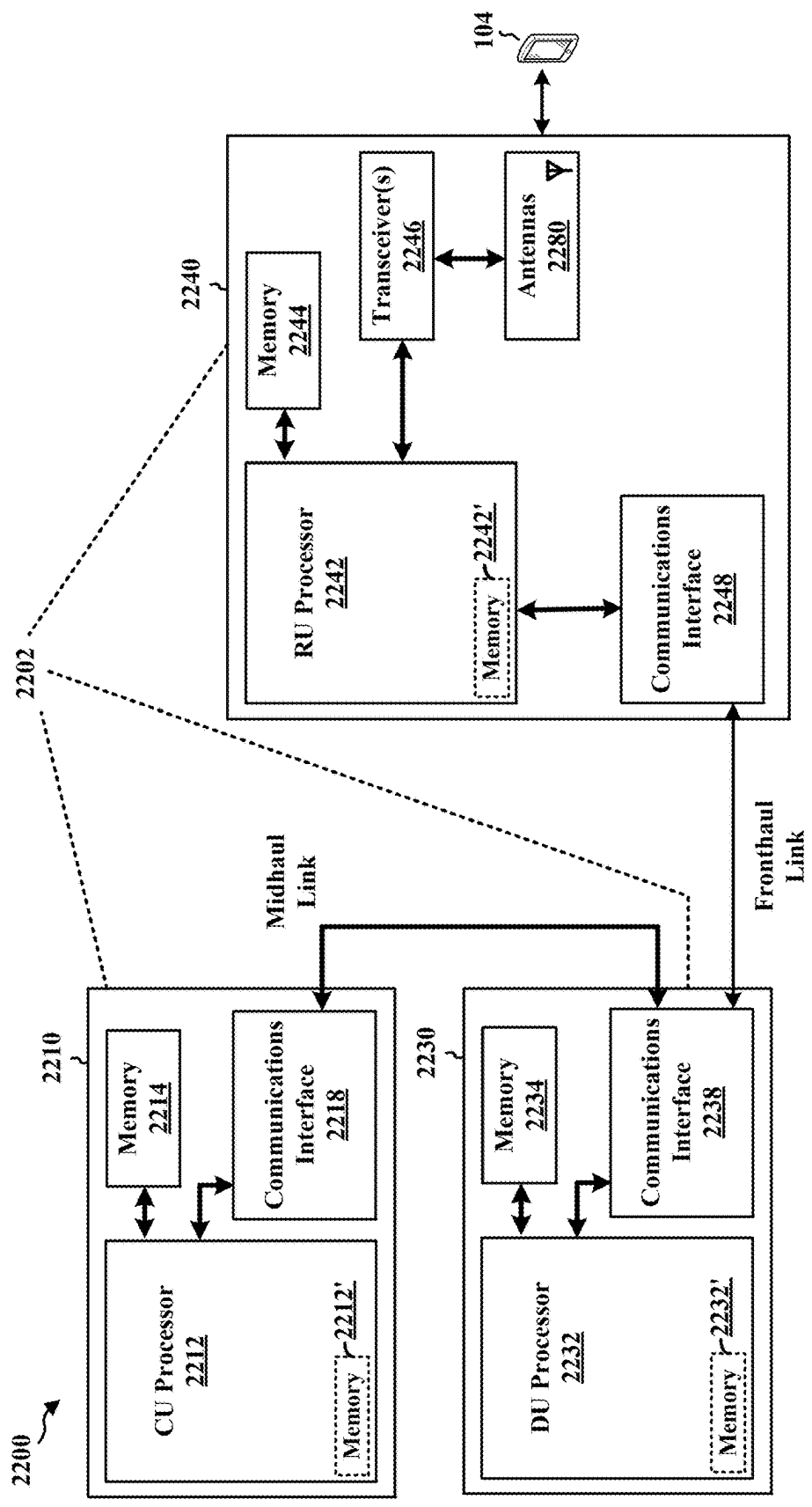


FIG. 22

DISCONTINUOUS RECEPTION WITH IMPLICIT INDICATION IN SIDELINK

TECHNICAL FIELD

[0001] The present disclosure relates generally to communication systems, and more particularly, to sidelink communications.

INTRODUCTION

[0002] Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, and broadcasts. Typical wireless communication systems may employ multiple-access technologies capable of supporting communication with multiple users by sharing available system resources. Examples of such multiple-access technologies include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, orthogonal frequency division multiple access (OFDMA) systems, single-carrier frequency division multiple access (SC-FDMA) systems, and time division synchronous code division multiple access (TD-SCDMA) systems.

[0003] These multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different wireless devices to communicate on a municipal, national, regional, and even global level. An example telecommunication standard is 5G New Radio (NR). 5G NR is part of a continuous mobile broadband evolution promulgated by Third Generation Partnership Project (3GPP) to meet new requirements associated with latency, reliability, security, scalability (e.g., with Internet of Things (IoT)), and other requirements. 5G NR includes services associated with enhanced mobile broadband (eMBB), massive machine type communications (mMTC), and ultra-reliable low latency communications (URLLC). Some aspects of 5G NR may be based on the 4G Long Term Evolution (LTE) standard. There exists a need for further improvements in 5G NR technology. These improvements may also be applicable to other multi-access technologies and the telecommunication standards that employ these technologies.

BRIEF SUMMARY

[0004] The following presents a simplified summary of one or more aspects in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated aspects. This summary neither identifies key or critical elements of all aspects nor delineates the scope of any or all aspects. Its sole purpose is to present some concepts of one or more aspects in a simplified form as a prelude to the more detailed description that is presented later.

[0005] In an aspect of the disclosure, a method, a computer-readable medium, and an apparatus for wireless communication at a first user equipment (UE) are provided. The apparatus includes a memory and at least one processor coupled to the memory and, based at least in part on information stored in the memory, the at least one processor is configured to: obtain a configuration associated with a discontinuous reception (DRX) cycle of a second UE, where the configuration includes a first indication of a first set of resources and at least one condition, where the first set of

resources is different from a second set of resources utilized for the DRX cycle of the second UE; and transmit, to the second UE based on an occurrence of the at least one condition, sidelink (SL) data via the first set of resources or the second set of resources.

[0006] In an aspect of the disclosure, a method, a computer-readable medium, and an apparatus for wireless communication at a second user equipment (UE) are provided. The apparatus includes a memory and at least one processor coupled to the memory and, based at least in part on information stored in the memory, the at least one processor is configured to: monitor for sidelink (SL) data via a first set of resources, where the first set of resources is different from a second set of resources utilized for a discontinuous reception (DRX) cycle of the second UE; and receive, from a first UE based on an occurrence of at least one condition, the SL data via the first set of resources or the second set of resources.

[0007] To the accomplishment of the foregoing and related ends, the one or more aspects comprise the features hereinafter fully described and particularly pointed out in the claims. The following description and the drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a diagram illustrating an example of a wireless communications system and an access network.

[0009] FIG. 2A is a diagram illustrating an example of a first frame, in accordance with various aspects of the present disclosure.

[0010] FIG. 2B is a diagram illustrating an example of downlink (DL) channels within a subframe, in accordance with various aspects of the present disclosure.

[0011] FIG. 2C is a diagram illustrating an example of a second frame, in accordance with various aspects of the present disclosure.

[0012] FIG. 2D is a diagram illustrating an example of uplink (UL) channels within a subframe, in accordance with various aspects of the present disclosure.

[0013] FIG. 3 is a diagram illustrating an example of a base station and user equipment (UE) in an access network.

[0014] FIG. 4 is a diagram illustrating an example of a discontinuous reception (DRX) cycle.

[0015] FIG. 5 is a diagram illustrating example communications between a first base station, a transmitting UE (Tx UE), a receiving UE (Rx UE), and a second base station for a sidelink DRX configuration.

[0016] FIG. 6 is a diagram illustrating example communications between a Tx UE and a Rx UE for a sidelink DRX configuration.

[0017] FIG. 7 is a diagram illustrating various example aspects of listen-before-talk (LBT) procedures.

[0018] FIG. 8 is a diagram illustrating examples of first stage sidelink control information (SCI-1) and second stage sidelink control information (SCI-2).

[0019] FIG. 9 is a diagram illustrating an example of a DRX on-duration and an extended DRX on-duration.

[0020] FIG. 10 is a diagram illustrating examples of an offset, a duration, a cycle length, and a cycle timer associated with an extended DRX on-duration.

[0021] FIG. 11 is a diagram illustrating an example of a fixed length extended DRX on-duration.

[0022] FIG. 12 is a diagram illustrating an example of a dynamic length extended DRX on-duration.

[0023] FIG. 13 is a diagram illustrating examples of extended DRX on-duration implicit indications.

[0024] FIG. 14 is a diagram illustrating an example of utilizing a fixed length extended DRX on-duration upon detection of sidelink control information (SCI).

[0025] FIG. 15 is a diagram illustrating an example of utilizing a dynamic length extended DRX on-duration upon detection of SCI.

[0026] FIG. 16 is a diagram illustrating example communications between a first UE and a second UE.

[0027] FIG. 17 is a flowchart of a method of wireless communication.

[0028] FIG. 18 is a flowchart of a method of wireless communication.

[0029] FIG. 19 is a flowchart of a method of wireless communication.

[0030] FIG. 20 is a flowchart of a method of wireless communication.

[0031] FIG. 21 is a diagram illustrating an example of a hardware implementation for an example apparatus.

[0032] FIG. 22 is a diagram illustrating an example of a hardware implementation for an example network entity.

DETAILED DESCRIPTION

[0033] A first UE may be configured with a discontinuous reception (DRX) cycle in which the first UE may receive packets during a DRX on-duration and in which the first UE may periodically enter a power-saving mode when there is no intimation of packet arrival from a base station. The first UE may also utilize a DRX cycle for sidelink communications with a second UE. When a DRX cycle is utilized in sidelink, a Rx UE (e.g., the first UE) may apply a restriction such that resources are within the DRX on-duration. If none of the resources are within the DRX on-duration of the Rx UE, an implementation of the Rx UE may determine whether resources are added to the DRX on-duration. Furthermore, in sidelink unlicensed (SL-U) communications, a Tx UE (e.g., the second UE) may perform a LBT procedure prior to transmission. If the LBT procedure fails, the Tx UE may issue a resource reselection to the Rx UE or the Tx UE may use a retransmission occasion for an initial transmission. Resource selection based on UE implementation may suffer from interference and may increase a probability of a subsequent LBT failure. Additionally, frequent resource reselection and/or insufficient resources may impede throughput. Various technologies pertaining to enhancing a sidelink DRX cycle are described herein. In an example, a first UE obtains a configuration associated with a DRX cycle of a second UE, where the configuration includes a first indication of a first set of resources (e.g., resources associated with a DRX on-duration of the second UE) and at least one condition, where the first set of resources is different from a second set of resources (e.g., resources associated with an extension of the DRX on-duration) utilized for the DRX cycle of the second UE. The first UE transmits, to the second UE based on an occurrence of the at least one condition, SL data via the first set of resources or the second set of resources. Vis-à-vis the configuration, the above-described technologies may increase communications reliability between the first UE and the second UE. For instance,

the configuration may reduce occurrences of resource reselection at a MAC layer and/or may reduce use of retransmission occasions for initial transmissions. The configuration may also reduce a probability of a LBT failure via reduced resource reselection.

[0034] The detailed description set forth below in connection with the drawings describes various configurations and does not represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of various concepts. However, these concepts may be practiced without these specific details. In some instances, well known structures and components are shown in block diagram form in order to avoid obscuring such concepts.

[0035] Several aspects of telecommunication systems are presented with reference to various apparatus and methods. These apparatus and methods are described in the following detailed description and illustrated in the accompanying drawings by various blocks, components, circuits, processes, algorithms, etc. (collectively referred to as “elements”). These elements may be implemented using electronic hardware, computer software, or any combination thereof. Whether such elements are implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system.

[0036] By way of example, an element, or any portion of an element, or any combination of elements may be implemented as a “processing system” that includes one or more processors. Examples of processors include microprocessors, microcontrollers, graphics processing units (GPUs), central processing units (CPUs), application processors, digital signal processors (DSPs), reduced instruction set computing (RISC) processors, systems on a chip (SoC), baseband processors, field programmable gate arrays (FPGAs), programmable logic devices (PLDs), state machines, gated logic, discrete hardware circuits, and other suitable hardware configured to perform the various functionality described throughout this disclosure. One or more processors in the processing system may execute software. Software, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise, shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software components, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, functions, or any combination thereof.

[0037] Accordingly, in one or more example aspects, implementations, and/or use cases, the functions described may be implemented in hardware, software, or any combination thereof. If implemented in software, the functions may be stored on or encoded as one or more instructions or code on a computer-readable medium. Computer-readable media includes computer storage media. Storage media may be any available media that can be accessed by a computer. By way of example, such computer-readable media can comprise a random-access memory (RAM), a read-only memory (ROM), an electrically erasable programmable ROM (EEPROM), optical disk storage, magnetic disk storage, other magnetic storage devices, combinations of the types of computer-readable media, or any other medium that

can be used to store computer executable code in the form of instructions or data structures that can be accessed by a computer.

[0038] While aspects, implementations, and/or use cases are described in this application by illustration to some examples, additional or different aspects, implementations and/or use cases may come about in many different arrangements and scenarios. Aspects, implementations, and/or use cases described herein may be implemented across many differing platform types, devices, systems, shapes, sizes, and packaging arrangements. For example, aspects, implementations, and/or use cases may come about via integrated chip implementations and other non-module-component based devices (e.g., end-user devices, vehicles, communication devices, computing devices, industrial equipment, retail/purchasing devices, medical devices, artificial intelligence (AI)-enabled devices, etc.). While some examples may or may not be specifically directed to use cases or applications, a wide assortment of applicability of described examples may occur. Aspects, implementations, and/or use cases may range a spectrum from chip-level or modular components to non-modular, non-chip-level implementations and further to aggregate, distributed, or original equipment manufacturer (OEM) devices or systems incorporating one or more techniques herein. In some practical settings, devices incorporating described aspects and features may also include additional components and features for implementation and practice of claimed and described aspect. For example, transmission and reception of wireless signals necessarily includes a number of components for analog and digital purposes (e.g., hardware components including antenna, RF-chains, power amplifiers, modulators, buffer, processor (s), interleaver, adders/summers, etc.). Techniques described herein may be practiced in a wide variety of devices, chip-level components, systems, distributed arrangements, aggregated or disaggregated components, end-user devices, etc. of varying sizes, shapes, and constitution.

[0039] Deployment of communication systems, such as 5G NR systems, may be arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a radio access network (RAN) node, a core network node, a network element, or a network equipment, such as a base station (BS), or one or more units (or one or more components) performing base station functionality, may be implemented in an aggregated or disaggregated architecture. For example, a BS (such as a Node B (NB), evolved NB (eNB), NR BS, 5G NB (e.g., a gNB), access point (AP), a transmit receive point (TRP), or a cell, etc.) may be implemented as an aggregated base station (also known as a standalone BS or a monolithic BS) or a disaggregated base station.

[0040] An aggregated base station may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node. A disaggregated base station may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more central or centralized units (CUs), one or more distributed units (DUs), or one or more radio units (RUs)). In some aspects, a CU may be implemented within a RAN node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other RAN nodes. The DUs may be implemented to communicate with one or

more RUs. Each of the CU, DU and RU can be implemented as virtual units, i.e., a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU).

[0041] Base station operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized in an integrated access backhaul (IAB) network, an open radio access network (O-RAN (such as the network configuration sponsored by the O-RAN Alliance)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)). Disaggregation may include distributing functionality across two or more units at various physical locations, as well as distributing functionality for at least one unit virtually, which can enable flexibility in network design. The various units of the disaggregated base station, or disaggregated RAN architecture, can be configured for wired or wireless communication with at least one other unit.

[0042] FIG. 1 is a diagram 100 illustrating an example of a wireless communications system and an access network. The illustrated wireless communications system includes a disaggregated base station architecture. The disaggregated base station architecture may include one or more CUs 110 that can communicate directly with a core network 120 via a backhaul link, or indirectly with the core network 120 through one or more disaggregated base station units (such as a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC) 125 via an E2 link, or a Non-Real Time (Non-RT) RIC 115 associated with a Service Management and Orchestration (SMO) Framework 105, or both). A CU 110 may communicate with one or more DUs 130 via respective midhaul links, such as an F1 interface. The DUs 130 may communicate with one or more RUs 140 via respective fronthaul links. The RUs 140 may communicate with respective UEs 104 via one or more radio frequency (RF) access links. In some implementations, the UE 104 may be simultaneously served by multiple RUs 140.

[0043] Each of the units, i.e., the CUs 110, the DUs 130, the RUs 140, as well as the Near-RT RICs 125, the Non-RT RICs 115, and the SMO Framework 105, may include one or more interfaces or be coupled to one or more interfaces configured to receive or to transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to the communication interfaces of the units, can be configured to communicate with one or more of the other units via the transmission medium. For example, the units can include a wired interface configured to receive or to transmit signals over a wired transmission medium to one or more of the other units. Additionally, the units can include a wireless interface, which may include a receiver, a transmitter, or a transceiver (such as an RF transceiver), configured to receive or to transmit signals, or both, over a wireless transmission medium to one or more of the other units.

[0044] In some aspects, the CU 110 may host one or more higher layer control functions. Such control functions can include radio resource control (RRC), packet data convergence protocol (PDCP), service data adaptation protocol (SDAP), or the like. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU 110. The CU 110 may be configured to handle user plane functionality (i.e., Central Unit-User Plane (CU-UP)), control plane func-

tionality (i.e., Central Unit-Control Plane (CU-CP)), or a combination thereof. In some implementations, the CU **110** can be logically split into one or more CU-UP units and one or more CU-CP units. The CU-UP unit can communicate bidirectionally with the CU-CP unit via an interface, such as an E1 interface when implemented in an O-RAN configuration. The CU **110** can be implemented to communicate with the DU **130**, as necessary, for network control and signaling.

[0045] The DU **130** may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs **140**. In some aspects, the DU **130** may host one or more of a radio link control (RLC) layer, a medium access control (MAC) layer, and one or more high physical (PHY) layers (such as modules for forward error correction (FEC) encoding and decoding, scrambling, modulation, demodulation, or the like) depending, at least in part, on a functional split, such as those defined by 3GPP. In some aspects, the DU **130** may further host one or more low PHY layers. Each layer (or module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU **130**, or with the control functions hosted by the CU **110**.

[0046] Lower-layer functionality can be implemented by one or more RUs **140**. In some deployments, an RU **140**, controlled by a DU **130**, may correspond to a logical node that hosts RF processing functions, or low-PHY layer functions (such as performing fast Fourier transform (FFT), inverse FFT (IFFT), digital beamforming, physical random access channel (PRACH) extraction and filtering, or the like), or both, based at least in part on the functional split, such as a lower layer functional split. In such an architecture, the RU(s) **140** can be implemented to handle over the air (OTA) communication with one or more UEs **104**. In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU(s) **140** can be controlled by the corresponding DU **130**. In some scenarios, this configuration can enable the DU(s) **130** and the CU **110** to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

[0047] The SMO Framework **105** may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework **105** may be configured to support the deployment of dedicated physical resources for RAN coverage requirements that may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework **105** may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) **190**) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs **110**, DUs **130**, RUs **140** and Near-RT RICs **125**. In some implementations, the SMO Framework **105** can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) **111**, via an O1 interface. Additionally, in some implementations, the SMO Framework **105** can communicate directly with one or more RUs **140** via an O1 interface. The SMO Framework **105** also may include a Non-RT RIC **115** configured to support functionality of the SMO Framework **105**.

[0048] The Non-RT RIC **115** may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, artificial intelligence (AI)/machine learning (ML) (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC **125**. The Non-RT RIC **115** may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC **125**. The Near-RT RIC **125** may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs **110**, one or more DUs **130**, or both, as well as an O-eNB, with the Near-RT RIC **125**.

[0049] In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC **125**, the Non-RT RIC **115** may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC **125** and may be received at the SMO Framework **105** or the Non-RT RIC **115** from non-network data sources or from network functions. In some examples, the Non-RT RIC **115** or the Near-RT RIC **125** may be configured to tune RAN behavior or performance. For example, the Non-RT RIC **115** may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework **105** (such as reconfiguration via **01**) or via creation of RAN management policies (such as A1 policies).

[0050] At least one of the CU **110**, the DU **130**, and the RU **140** may be referred to as a base station **102**. Accordingly, a base station **102** may include one or more of the CU **110**, the DU **130**, and the RU **140** (each component indicated with dotted lines to signify that each component may or may not be included in the base station **102**). The base station **102** provides an access point to the core network **120** for a UE **104**. The base stations **102** may include macrocells (high power cellular base station) and/or small cells (low power cellular base station). The small cells include femtocells, picocells, and microcells. A network that includes both small cell and macrocells may be known as a heterogeneous network. A heterogeneous network may also include Home Evolved Node Bs (eNBs) (HeNBs), which may provide service to a restricted group known as a closed subscriber group (CSG). The communication links between the RUs **140** and the UEs **104** may include uplink (UL) (also referred to as reverse link) transmissions from a UE **104** to an RU **140** and/or downlink (DL) (also referred to as forward link) transmissions from an RU **140** to a UE **104**. The communication links may use multiple-input and multiple-output (MIMO) antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links may be through one or more carriers. The base stations **102**/UEs **104** may use spectrum up to Y MHz (e.g., 5, 10, 15, 20, 100, 400, etc. MHz) bandwidth per carrier allocated in a carrier aggregation of up to a total of Yx MHz (x component carriers) used for transmission in each direction. The carriers may or may not be adjacent to each other. Allocation of carriers may be asymmetric with respect to DL and UL (e.g., more or fewer carriers may be allocated for DL than for UL). The component carriers may include a primary component carrier and one or more secondary component carriers. A primary component carrier may be referred to as a primary cell (PCell) and a secondary component carrier may be referred to as a secondary cell (SCell).

[0051] Certain UEs **104** may communicate with each other using device-to-device (D2D) communication link **158**. The D2D communication link **158** may use the DL/UL wireless wide area network (WWAN) spectrum. The D2D communication link **158** may use one or more sidelink channels, such as a physical sidelink broadcast channel (PSBCH), a physical sidelink discovery channel (PSDCH), a physical sidelink shared channel (PSSCH), and a physical sidelink control channel (PSCCH). D2D communication may be through a variety of wireless D2D communications systems, such as for example, Bluetooth, Wi-Fi based on the Institute of Electrical and Electronics Engineers (IEEE) 802.11 standard, LTE, or NR.

[0052] The wireless communications system may further include a Wi-Fi AP **150** in communication with UEs **104** (also referred to as Wi-Fi stations (STAs)) via communication link **154**, e.g., in a 5 GHz unlicensed frequency spectrum or the like. When communicating in an unlicensed frequency spectrum, the UEs **104**/AP **150** may perform a clear channel assessment (CCA) prior to communicating in order to determine whether the channel is available.

[0053] The electromagnetic spectrum is often subdivided, based on frequency/wavelength, into various classes, bands, channels, etc. In 5G NR, two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25 GHz-52.6 GHz). Although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a “sub-6 GHz” band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a “millimeter wave” band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the International Telecommunications Union (ITU) as a “millimeter wave” band.

[0054] The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHz-24.25 GHz). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR2-2 (52.6 GHz-71 GHz), FR4 (71 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.

[0055] With the above aspects in mind, unless specifically stated otherwise, the term “sub-6 GHz” or the like if used herein may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, the term “millimeter wave” or the like if used herein may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR2-2, and/or FR5, or may be within the EHF band.

[0056] The base station **102** and the UE **104** may each include a plurality of antennas, such as antenna elements, antenna panels, and/or antenna arrays to facilitate beamforming. The base station **102** may transmit a beamformed signal **182** to the UE **104** in one or more transmit directions. The UE **104** may receive the beamformed signal from the

base station **102** in one or more receive directions. The UE **104** may also transmit a beamformed signal **184** to the base station **102** in one or more transmit directions. The base station **102** may receive the beamformed signal from the UE **104** in one or more receive directions. The base station **102**/UE **104** may perform beam training to determine the best receive and transmit directions for each of the base station **102**/UE **104**. The transmit and receive directions for the base station **102** may or may not be the same. The transmit and receive directions for the UE **104** may or may not be the same.

[0057] The base station **102** may include and/or be referred to as a gNB, Node B, eNB, an access point, a base transceiver station, a radio base station, a radio transceiver, a transceiver function, a basic service set (BSS), an extended service set (ESS), a transmit reception point (TRP), network node, network entity, network equipment, or some other suitable terminology. The base station **102** can be implemented as an integrated access and backhaul (IAB) node, a relay node, a sidelink node, an aggregated (monolithic) base station with a baseband unit (BBU) (including a CU and a DU) and an RU, or as a disaggregated base station including one or more of a CU, a DU, and/or an RU. The set of base stations, which may include disaggregated base stations and/or aggregated base stations, may be referred to as next generation (NG) RAN (NG-RAN).

[0058] The core network **120** may include an Access and Mobility Management Function (AMF) **161**, a Session Management Function (SMF) **162**, a User Plane Function (UPF) **163**, a Unified Data Management (UDM) **164**, one or more location servers **168**, and other functional entities. The AMF **161** is the control node that processes the signaling between the UEs **104** and the core network **120**. The AMF **161** supports registration management, connection management, mobility management, and other functions. The SMF **162** supports session management and other functions. The UPF **163** supports packet routing, packet forwarding, and other functions. The UDM **164** supports the generation of authentication and key agreement (AKA) credentials, user identification handling, access authorization, and subscription management. The one or more location servers **168** are illustrated as including a Gateway Mobile Location Center (GMLC) **165** and a Location Management Function (LMF) **166**. However, generally, the one or more location servers **168** may include one or more location/positioning servers, which may include one or more of the GMLC **165**, the LMF **166**, a position determination entity (PDE), a serving mobile location center (SMLC), a mobile positioning center (MPC), or the like. The GMLC **165** and the LMF **166** support UE location services. The GMLC **165** provides an interface for clients/applications (e.g., emergency services) for accessing UE positioning information. The LMF **166** receives measurements and assistance information from the NG-RAN and the UE **104** via the AMF **161** to compute the position of the UE **104**. The NG-RAN may utilize one or more positioning methods in order to determine the position of the UE **104**. Positioning the UE **104** may involve signal measurements, a position estimate, and an optional velocity computation based on the measurements. The signal measurements may be made by the UE **104** and/or the serving base station **102**. The signals measured may be based on one or more of a satellite positioning system (SPS) **170** (e.g., one or more of a Global Navigation Satellite System (GNSS), global position system (GPS), non-terrestrial network

(NTN), or other satellite position/location system), LTE signals, wireless local area network (WLAN) signals, Bluetooth signals, a terrestrial beacon system (TBS), sensor-based information (e.g., barometric pressure sensor, motion sensor), NR enhanced cell ID (NR E-CID) methods, NR signals (e.g., multi-round trip time (Multi-RTT), DL angle-of-departure (DL-AoD), DL time difference of arrival (DL-TDOA), UL time difference of arrival (UL-TDOA), and UL angle-of-arrival (UL-AoA) positioning), and/or other systems/signals/sensors.

[0059] Examples of UEs **104** include a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a laptop, a personal digital assistant (PDA), a satellite radio, a global positioning system, a multimedia device, a video device, a digital audio player (e.g., MP3 player), a camera, a game console, a tablet, a smart device, a wearable device, a vehicle, an electric meter, a gas pump, a large or small kitchen appliance, a healthcare device, an implant, a sensor/actuator, a display, or any other similar functioning device. Some of the UEs **104** may be referred to as IoT devices (e.g., parking meter, gas pump, toaster, vehicles, heart monitor, etc.). The UE **104** may also be referred to as a station, a mobile station, a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a mobile device, a wireless device, a wireless communications device, a remote device, a mobile subscriber station, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, a user agent, a mobile client, a client, or some other suitable terminology. In some scenarios, the term UE may also apply to one or more companion devices such as in a device constellation arrangement. One or more of these devices may collectively access the network and/or individually access the network.

[0060] Referring again to FIG. 1, in certain aspects, the UE **104** may include a DRX component **198** that is configured to obtain a configuration associated with a DRX cycle of a second UE, where the configuration includes a first indication of a first set of resources and at least one condition, where the first set of resources is different from a second set of resources utilized for the DRX cycle of the second UE; and transmit, to the second UE based on an occurrence of the at least one condition, SL data via the first set of resources or the second set of resources. In certain aspects, the DRX component **198** is configured to monitor for SL data via a first set of resources, where the first set of resources is different from a second set of resources utilized for a DRX cycle of the second UE; and receive, from a first UE based on an occurrence of at least one condition, the SL data via the first set of resources or the second set of resources. Although the following description may be focused on 5G NR, the concepts described herein may be applicable to other similar areas, such as LTE, LTE-A, CDMA, GSM, and other wireless technologies.

[0061] FIG. 2A is a diagram **200** illustrating an example of a first subframe within a 5G NR frame structure. FIG. 2B is a diagram **230** illustrating an example of DL channels within a 5G NR subframe. FIG. 2C is a diagram **250** illustrating an example of a second subframe within a 5G NR frame structure. FIG. 2D is a diagram **280** illustrating an example of UL channels within a 5G NR subframe. The 5G NR frame structure may be frequency division duplexed (FDD) in which for a particular set of subcarriers (carrier system bandwidth), subframes within the set of subcarriers are dedicated for either DL or UL, or may be time division

duplexed (TDD) in which for a particular set of subcarriers (carrier system bandwidth), subframes within the set of subcarriers are dedicated for both DL and UL. In the examples provided by FIGS. 2A, 2C, the 5G NR frame structure is assumed to be TDD, with subframe 4 being configured with slot format 28 (with mostly DL), where D is DL, U is UL, and F is flexible for use between DL/UL, and subframe 3 being configured with slot format 1 (with all UL). While subframes 3, 4 are shown with slot formats 1, 28, respectively, any particular subframe may be configured with any of the various available slot formats 0-61. Slot formats 0, 1 are all DL, UL, respectively. Other slot formats 2-61 include a mix of DL, UL, and flexible symbols. UEs are configured with the slot format (dynamically through DL control information (DCI), or semi-statically/statically through radio resource control (RRC) signaling) through a received slot format indicator (SFI). Note that the description infra applies also to a 5G NR frame structure that is TDD.

[0062] FIGS. 2A-2D illustrate a frame structure, and the aspects of the present disclosure may be applicable to other wireless communication technologies, which may have a different frame structure and/or different channels. A frame (10 ms) may be divided into 10 equally sized subframes (1 ms). Each subframe may include one or more time slots. Subframes may also include mini-slots, which may include 7, 4, or 2 symbols. Each slot may include 14 or 12 symbols, depending on whether the cyclic prefix (CP) is normal or extended. For normal CP, each slot may include 14 symbols, and for extended CP, each slot may include 12 symbols. The symbols on DL may be CP orthogonal frequency division multiplexing (OFDM) (CP-OFDM) symbols. The symbols on UL may be CP-OFDM symbols (for high throughput scenarios) or discrete Fourier transform (DFT) spread OFDM (DFT-s-OFDM) symbols (for power limited scenarios; limited to a single stream transmission). The number of slots within a subframe is based on the CP and the numerology. The numerology defines the subcarrier spacing (SCS) (see Table 1). The symbol length/duration may scale with $1/\text{SCS}$.

TABLE 1

Numerology, SCS, and CP		
μ	SCS $\Delta f = 2^\mu \cdot 15[\text{kHz}]$	Cyclic prefix
0	15	Normal
1	30	Normal
2	60	Normal, Extended
3	120	Normal
4	240	Normal
5	480	Normal
6	960	Normal

[0063] For normal CP (14 symbols/slot), different numerologies μ 0 to 4 allow for 1, 2, 4, 8, and 16 slots, respectively, per subframe. For extended CP, the numerology 2 allows for 4 slots per subframe. Accordingly, for normal CP and numerology μ , there are 14 symbols/slot and 2^μ slots/subframe. The subcarrier spacing may be equal to $2^\mu \cdot 15$ kHz, where μ is the numerology 0 to 4. As such, the numerology $\mu=0$ has a subcarrier spacing of 15 kHz and the numerology $\mu=4$ has a subcarrier spacing of 240 kHz. The symbol length/duration is inversely related to the subcarrier

spacing. FIGS. 2A-2D provide an example of normal CP with 14 symbols per slot and numerology $\mu=2$ with 4 slots per subframe. The slot duration is 0.25 ms, the subcarrier spacing is 60 kHz, and the symbol duration is approximately 16.67 μ s. Within a set of frames, there may be one or more different bandwidth parts (BWPs) (see FIG. 2B) that are frequency division multiplexed. Each BWP may have a particular numerology and CP (normal or extended).

[0064] A resource grid may be used to represent the frame structure. Each time slot includes a resource block (RB) (also referred to as physical RBs (PRBs)) that extends 12 consecutive subcarriers. The resource grid is divided into multiple resource elements (REs). The number of bits carried by each RE depends on the modulation scheme.

[0065] As illustrated in FIG. 2A, some of the REs carry reference (pilot) signals (RS) for the UE. The RS may include demodulation RS (DM-RS) (indicated as R for one particular configuration, but other DM-RS configurations are possible) and channel state information reference signals (CSI-RS) for channel estimation at the UE. The RS may also include beam measurement RS (BRS), beam refinement RS (BRRS), and phase tracking RS (PT-RS).

[0066] FIG. 2B illustrates an example of various DL channels within a subframe of a frame. The physical downlink control channel (PDCCH) carries DCI within one or more control channel elements (CCEs) (e.g., 1, 2, 4, 8, or 16 CCEs), each CCE including six RE groups (REGs), each REG including 12 consecutive REs in an OFDM symbol of an RB. A PDCCH within one BWP may be referred to as a control resource set (CORESET). A UE is configured to monitor PDCCH candidates in a PDCCH search space (e.g., common search space, UE-specific search space) during PDCCH monitoring occasions on the CORESET, where the PDCCH candidates have different DCI formats and different aggregation levels. Additional BWPs may be located at greater and/or lower frequencies across the channel bandwidth. A primary synchronization signal (PSS) may be within symbol 2 of particular subframes of a frame. The PSS is used by a UE 104 to determine subframe/symbol timing and a physical layer identity. A secondary synchronization signal (SSS) may be within symbol 4 of particular subframes of a frame. The SSS is used by a UE to determine a physical layer cell identity group number and radio frame timing. Based on the physical layer identity and the physical layer cell identity group number, the UE can determine a physical cell identifier (PCI). Based on the PCI, the UE can determine the locations of the DM-RS. The physical broadcast channel (PBCH), which carries a master information block (MIB), may be logically grouped with the PSS and SSS to form a synchronization signal (SS)/PBCH block (also referred to as SS block (SSB)). The MIB provides a number of RBs in the system bandwidth and a system frame number (SFN). The physical downlink shared channel (PDSCH) carries user data, broadcast system information not transmitted through the PBCH such as system information blocks (SIBs), and paging messages.

[0067] As illustrated in FIG. 2C, some of the REs carry DM-RS (indicated as R for one particular configuration, but other DM-RS configurations are possible) for channel estimation at the base station. The UE may transmit DM-RS for the physical uplink control channel (PUCCH) and DM-RS for the physical uplink shared channel (PUSCH). The PUSCH DM-RS may be transmitted in the first one or two symbols of the PUSCH. The PUCCH DM-RS may be

transmitted in different configurations depending on whether short or long PUCCHs are transmitted and depending on the particular PUCCH format used. The UE may transmit sounding reference signals (SRS). The SRS may be transmitted in the last symbol of a subframe. The SRS may have a comb structure, and a UE may transmit SRS on one of the combs. The SRS may be used by a base station for channel quality estimation to enable frequency-dependent scheduling on the UL.

[0068] FIG. 2D illustrates an example of various UL channels within a subframe of a frame. The PUCCH may be located as indicated in one configuration. The PUCCH carries uplink control information (UCI), such as scheduling requests, a channel quality indicator (CQI), a precoding matrix indicator (PMI), a rank indicator (RI), and hybrid automatic repeat request (HARQ) acknowledgment (ACK) (HARQ-ACK) feedback (i.e., one or more HARQ ACK bits indicating one or more ACK and/or negative ACK (NACK)). The PUSCH carries data, and may additionally be used to carry a buffer status report (BSR), a power headroom report (PHR), and/or UCI.

[0069] FIG. 3 is a block diagram of a base station 310 in communication with a UE 350 in an access network. In the DL, Internet protocol (IP) packets may be provided to a controller/processor 375. The controller/processor 375 implements layer 3 and layer 2 functionality. Layer 3 includes a radio resource control (RRC) layer, and layer 2 includes a service data adaptation protocol (SDAP) layer, a packet data convergence protocol (PDCP) layer, a radio link control (RLC) layer, and a medium access control (MAC) layer. The controller/processor 375 provides RRC layer functionality associated with broadcasting of system information (e.g., MIB, SIBs), RRC connection control (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), inter radio access technology (RAT) mobility, and measurement configuration for UE measurement reporting; PDCP layer functionality associated with header compression/decompression, security (ciphering, deciphering, integrity protection, integrity verification), and handover support functions; RLC layer functionality associated with the transfer of upper layer packet data units (PDUs), error correction through ARQ, concatenation, segmentation, and reassembly of RLC service data units (SDUs), re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto transport blocks (TBs), demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through HARQ, priority handling, and logical channel prioritization.

[0070] The transmit (TX) processor 316 and the receive (RX) processor 370 implement layer 1 functionality associated with various signal processing functions. Layer 1, which includes a physical (PHY) layer, may include error detection on the transport channels, forward error correction (FEC) coding/decoding of the transport channels, interleaving, rate matching, mapping onto physical channels, modulation/demodulation of physical channels, and MIMO antenna processing. The TX processor 316 handles mapping to signal constellations based on various modulation schemes (e.g., binary phase-shift keying (BPSK), quadrature phase-shift keying (QPSK), M-phase-shift keying (M-PSK), M-quadrature amplitude modulation (M-QAM)). The coded

and modulated symbols may then be split into parallel streams. Each stream may then be mapped to an OFDM subcarrier, multiplexed with a reference signal (e.g., pilot) in the time and/or frequency domain, and then combined together using an Inverse Fast Fourier Transform (IFFT) to produce a physical channel carrying a time domain OFDM symbol stream. The OFDM stream is spatially precoded to produce multiple spatial streams. Channel estimates from a channel estimator 374 may be used to determine the coding and modulation scheme, as well as for spatial processing. The channel estimate may be derived from a reference signal and/or channel condition feedback transmitted by the UE 350. Each spatial stream may then be provided to a different antenna 320 via a separate transmitter 318Tx. Each transmitter 318Tx may modulate a radio frequency (RF) carrier with a respective spatial stream for transmission.

[0071] At the UE 350, each receiver 354Rx receives a signal through its respective antenna 352. Each receiver 354Rx recovers information modulated onto an RF carrier and provides the information to the receive (RX) processor 356. The TX processor 368 and the RX processor 356 implement layer 1 functionality associated with various signal processing functions. The RX processor 356 may perform spatial processing on the information to recover any spatial streams destined for the UE 350. If multiple spatial streams are destined for the UE 350, they may be combined by the RX processor 356 into a single OFDM symbol stream. The RX processor 356 then converts the OFDM symbol stream from the time-domain to the frequency domain using a Fast Fourier Transform (FFT). The frequency domain signal comprises a separate OFDM symbol stream for each subcarrier of the OFDM signal. The symbols on each subcarrier, and the reference signal, are recovered and demodulated by determining the most likely signal constellation points transmitted by the base station 310. These soft decisions may be based on channel estimates computed by the channel estimator 358. The soft decisions are then decoded and deinterleaved to recover the data and control signals that were originally transmitted by the base station 310 on the physical channel. The data and control signals are then provided to the controller/processor 359, which implements layer 3 and layer 2 functionality.

[0072] The controller/processor 359 can be associated with a memory 360 that stores program codes and data. The memory 360 may be referred to as a computer-readable medium. In the UL, the controller/processor 359 provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, and control signal processing to recover IP packets. The controller/processor 359 is also responsible for error detection using an ACK and/or NACK protocol to support HARQ operations.

[0073] Similar to the functionality described in connection with the DL transmission by the base station 310, the controller/processor 359 provides RRC layer functionality associated with system information (e.g., MIB, SIBs) acquisition, RRC connections, and measurement reporting; PDCP layer functionality associated with header compression/decompression, and security (ciphering, deciphering, integrity protection, integrity verification); RLC layer functionality associated with the transfer of upper layer PDUs, error correction through ARQ, concatenation, segmentation, and reassembly of RLC SDUs, re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer

functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto TBs, demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through HARQ, priority handling, and logical channel prioritization.

[0074] Channel estimates derived by a channel estimator 358 from a reference signal or feedback transmitted by the base station 310 may be used by the TX processor 368 to select the appropriate coding and modulation schemes, and to facilitate spatial processing. The spatial streams generated by the TX processor 368 may be provided to different antenna 352 via separate transmitters 354Tx. Each transmitter 354Tx may modulate an RF carrier with a respective spatial stream for transmission.

[0075] The UL transmission is processed at the base station 310 in a manner similar to that described in connection with the receiver function at the UE 350. Each receiver 318Rx receives a signal through its respective antenna 320. Each receiver 318Rx recovers information modulated onto an RF carrier and provides the information to a RX processor 370.

[0076] The controller/processor 375 can be associated with a memory 376 that stores program codes and data. The memory 376 may be referred to as a computer-readable medium. In the UL, the controller/processor 375 provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, control signal processing to recover IP packets. The controller/processor 375 is also responsible for error detection using an ACK and/or NACK protocol to support HARQ operations.

[0077] At least one of the TX processor 368, the RX processor 356, and the controller/processor 359 may be configured to perform aspects in connection with the DRX component 198 of FIG. 1.

[0078] FIG. 4 is a diagram 400 illustrating an example of a DRX cycle. When a UE is configured with a DRX cycle, the UE may not continually monitor a PDCCH for a PDCCH transmission. A DRX cycle may be characterized by an on-duration and an inactivity timer. The on-duration may refer to a duration that a UE waits for after waking up to receive PDCCH transmissions. If the UE successfully decodes a PDCCH transmission during an on-duration, the UE may stay awake and start the inactivity timer. The inactivity timer may refer to a duration that the UE waits to successfully decode a PDCCH transmission from a last successful decoding of a prior PDCCH transmission. If the UE does not successfully decode a PDCCH transmission during the duration of the inactivity timer, the UE may be placed in a sleep state. The UE may restart the inactivity timer following a single successfully decoding of a PDCCH transmission. The UE may not restart the inactivity timer for a retransmission of a PDCCH transmission. If a SL UE is configured with a DRX cycle, a PDCCH providing SL grants may be sent to the UE during an active time of the UE.

[0079] A UE may be configured with a (long) DRX cycle 404. During a long DRX on-duration 402 of the (long) DRX cycle 404, the UE may monitor for data (e.g., downlink data, such as downlink control signaling, or sidelink data, such as sidelink control signaling). In an example, the UE may monitor a PDCCH or a PSCCH for transmissions during the long DRX on-duration 402. If the UE successfully decodes a transmission (e.g., a PDCCH transmission or a PSCCH transmission) during the long DRX on-duration 402, the UE may stay awake and start an inactivity timer 405 (e.g.,

specified in milliseconds). When the UE is not within the long DRX on-duration 402 of the (long) DRX cycle 404 and if the UE does not successfully decode a transmission during a duration associated with the inactivity timer 405, the UE may sleep with receiver circuitry turned off. This may allow for a reduction in power consumption of the UE. In an example, a relatively longer DRX cycle may enable a relatively greater power consumption reduction than a relatively shorter DRX cycle.

[0080] The UE may also be configured with a short DRX cycle 406 in addition to the (long) DRX cycle 404. During a short DRX on-duration 403 of the short DRX cycle 406, a UE may monitor for a transmission (e.g., a PDCCH transmission or a PSCCH transmission). The short DRX cycle 406 may have a shorter duration than the (long) DRX cycle 404. In an example, the UE may follow the (long) DRX cycle 404 until the UE is scheduled for the short DRX cycle 406. In the example, the UE may follow the short DRX cycle 406 for a period of time (in addition to following the (long) DRX cycle 404). The (long) DRX cycle 404 may be associated with reduced power consumption; however, the (long) DRX cycle may be associated with increased latency for receiving data. When an inactivity timer expires (e.g., the inactivity timer 405), the short DRX cycle 406 may be activated. The short DRX cycle 406 may be associated with reduced latency for receiving data. The short DRX cycle 406 may be useful in voice over internet protocol (IP) related scenarios.

[0081] A UE may be configured with a connected mode DRX (C-DRX) mechanism that may enable a UE (e.g., a mobile UE) to periodically enter a power-saving mode (i.e., sleep mode). During the power-saving mode, the UE may turn off major circuits when there is no indication of packet arrival. The UE may wake up (i.e., leave the sleep mode) to periodically check for packet arrival. To prevent data loss, the UE and a network may have a predefined agreement about a periodic transition of the UE between sleep states and non-sleep states (i.e., wake up states). In an example, the UE may receive DRX configuration parameters in a DL RRC configuration message sent by the network (e.g., via a base station, such as a gNB).

[0082] FIG. 5 is a diagram 500 illustrating example communications between a first base station 502, a Tx UE 504, a Rx UE 506, and a second base station 508 for a sidelink DRX configuration (e.g., a C-DRX sidelink configuration). The communications in the diagram 500 may be associated with a sidelink DRX configuration that is unicast, that is, negotiated between UEs. In an example, the Tx UE 504 and/or the Rx UE 506 may be in a RRC connected state. At 510, the Rx UE 506 may transmit sidelink (SL) UE assistance information to the Tx UE 504. The SL UE assistance information may include an indication of one or more DRX cycle lengths, one or more DRX cycle offsets, and one or more DRX on-durations. Stated differently, the SL UE assistance information may be a desired SL DRX configuration.

[0083] At 512, the Tx UE 504 may forward the SL UE assistance information to the first base station 502 (e.g., a first gNB). At 514, the first base station 502 may determine a SL DRX configuration for the Rx UE 506 and the first base station 502 may transmit a SL DRX configuration to the Tx UE 504. The first base station 502 may determine the SL DRX configuration for the Rx UE 506 based on the SL UE assistance information. In one aspect, at 516, the first base

station 502 may align the SL DRX cycle with a DRX cycle of the Tx UE 504 (e.g., “align SL DRX with Tx UE UE-UTRAN (Uu) DRX”).

[0084] At 518, the Tx UE 504 may transmit the SL DRX configuration (i.e., a Rx UE SL DRX) to the Rx UE 506 via a PC-5 RRC message. PC-5 may refer to an interface used for sidelink communications. A PC-5 interface may support SL vehicle-to-everything (V2X) communications for NR and LTE. At 520, the Rx UE 506 may transmit a message to the Tx UE 504 indicating whether the SL DRX configuration is accepted or rejected. At 522, if the SL DRX configuration is accepted by the Rx UE 506, the Rx UE 506 may transmit the SL DRX configuration to the second base station 508 (e.g., a second gNB). In one aspect, at 524, the second base station 508 may align a DRX cycle of the Rx UE 506 with the SL DRX cycle (e.g., “align Rx UE Uu DRX with SL DRX”).

[0085] FIG. 6 is a diagram 600 illustrating example communications between the Tx UE 504 and the Rx UE 506 for a sidelink DRX configuration. The communications in the diagram 600 may be associated with a sidelink DRX configuration that is unicast, that is, negotiated between UEs. In an example, the Tx UE 504 and/or the Rx UE 506 may be in a RRC inactive state or a RRC idle state. In another example, the communications in the diagram 600 may be applicable when UEs are in coverage (IC) or out of coverage (OoC) of a base station.

[0086] At 602, the Rx UE 506 may transmit SL UE assistance information to the Tx UE 504. The SL UE assistance information may include an indication of one or more DRX cycle lengths, one or more DRX cycle offsets, and one or more DRX on-durations. Stated differently, the SL UE assistance information may be a desired SL DRX configuration. At 604, the Tx UE 504 may determine a SL DRX configuration for the Rx UE 506. The Tx UE 504 may determine the SL DRX configuration based on the SL UE assistance information. The Tx UE 504 may determine the SL DRX configuration based on an implementation of the Tx UE 504 and/or the Rx UE 506. In an example, an implementation of the Rx UE 506 may be utilized to derive an inactivity timer associated with the SL DRX configuration. At 606, the Tx UE 504 may transmit the SL DRX configuration (i.e., a Rx UE SL DRX) to the Rx UE 506 via a PC-5 RRC message. At 608, the Rx UE 506 may transmit a message to the Tx UE 504 indicating whether the SL DRX configuration is accepted or rejected.

[0087] FIG. 7 is a diagram 700 illustrating various example aspects associated with LBT procedures. The diagram 700 includes a first example 702 that depicts a LBT procedure. During the LBT procedure, a UE may refrain from transmitting data while a channel is unavailable 704, that is, the UE may monitor the channel and determine that the channel is unavailable based on the monitoring. The LBT procedure may be used for sidelink communications between UEs. The LBT procedure may be a Type 1 LBT procedure that may be used to initiate one or more transmission with the same channel occupancy time (COT).

[0088] When the UE determines that the channel is available for at least a defer period 706, the UE may initiate a backoff procedure 708. In an example, the defer period 706 may range from $16 \times 9 \times n$ ms. In an example, n may be 1, 3, or 7 for downlink and n may be 2, 3, or 7 for uplink. In an example, n may be specified in a downlink channel access priority class (CAPC) table or an uplink CAPC table. The

UE may determine that the channel is available if a received energy during the defer period is less than a threshold value. The backoff procedure **708** may include initializing a backoff counter with a random number within a contention window (CW). The random number may range from zero to the CW. The random number may represent a duration that the channel is to be available before data may be transmitted over the channel. The backoff counter may be decremented at certain intervals. Each time the backoff counter is decremented, the UE may determine whether or not the channel is idle for a time period (i.e., whether the channel is available). If the channel is idle, the backoff counter may be decremented again. If the channel is not idle, the UE may wait until the channel is idle before resuming the backoff counter. When the backoff counter reaches zero, data transmission **710** by the UE may occur.

[0089] The diagram **700** also includes a second example **712** that depicts an example of a COT **714**. Following a successful dynamic or semi-static channel access procedure, a channel may be used during the COT **714**. During the COT **714**, one or more transmission bursts (e.g., for uplink or downlink) may be exchanged between devices (e.g., between UEs).

[0090] The second example **712** depicts a first transmission burst **716** and a second transmission burst **718** that may be separated by a gap **720**. Depending on a size of the gap **720**, a UE may utilize a different type of channel access procedure (e.g., a different type of a type 2 LBT procedure). The second example **712** depicts a table **722** that details gap sizes and corresponding type 2 LBT procedures. If the size of the gap **720** is greater than or equal to 25 μ s, a device (e.g., a UE) may utilize a type 2A LBT procedure. In an example, a device (e.g., a base station, such as a gNB) may transmit a DL transmission immediately after sensing a channel to be idle for a sensing interval of 25 μ s. If the size of the gap **720** is greater than or equal to 16 μ s and less than 25 μ s, the device may utilize a cyclic prefix (CP) extension to maintain a 16 μ s gap and the device may utilize a type 2B LBT procedure. In an example, a device (e.g., a base station, such as a gNB) may transmit a DL transmission immediately after sensing a channel to be idle within a duration of 16 μ s. If the size of the gap **720** is less than or equal to 16 μ s, the device may utilize a type 2C LBT procedure. In an example, if a device (e.g., a base station, such as a gNB) follows a type 2C LBT procedure, the device may not sense the channel before transmission of a DL transmission. The DL transmission may have a duration that is less than or equal to 584 μ s.

[0091] In an example, a type 1 LBT procedure may be performed prior to transmission of the first transmission burst **716**. A type 2A, type 2B, or type 2C LBT procedure may be performed prior to transmission of the second transmission burst **718** based on a size of the gap **720**. In an example, the first transmission burst **716** may be a downlink transmission and the second transmission burst **718** may be an uplink transmission. In another example, the first transmission burst **716** and the second transmission burst **718** may be communications exchanged between UEs in sidelink communications. Furthermore, the type 2A, type 2B, and type 2C LBT procedures may be used for sidelink communications between UEs.

[0092] FIG. **8** is a diagram **800** illustrating examples of a SCI-1 **802** and a SCI-2 **804**. The SCI-1 **802** may be referred to as a first stage SCI and the SCI-2 **804** may be referred to as a second stage SCI. The SCI-1 **802** may also be referred

to as “SCI format 1” and the SCI-2 **804** may also be referred to as “SCI format 2.” In general, the SCI-1 **802** may include information that may be used by a UE to decode information in the SCI-2 **804**.

[0093] The SCI-1 **802** may be carried on a PSSCH that is associated with a PSSCH. The SCI-1 **802** may include information for demodulation/detection of the PSSCH. The SCI-1 **802** may include a priority indication **806**. The SCI-1 **802** may include an indication of a frequency resource assignment **808** for a UE. The SCI-1 **802** may include an indication of a time resource assignment **810**. The SCI-1 **802** may include an indication of a resource reservation period **812**. The SCI-1 **802** may include an indication of a DM-RS pattern **814**. The SCI-1 **802** may include an indication of a SCI-2 format **816** for a to-be-received SCI-2 (e.g., the SCI-2 **804**). In an example, the SCI-2 **804** may be in one of a plurality of formats and the SCI-2 format **816** may indicate such a format. The SCI-1 **802** may include a modulation and coding scheme **818**. The SCI-1 **802** may include a reserved portion **820**. The SCI-1 **802** may include a beta offset indicator **822**. The SCI-1 **802** may include a number of a DM-RS port **824**.

[0094] The SCI-2 **804** may be carried on a PSSCH. The SCI-2 **804** may include a HARQ process identifier (ID) **826**. The SCI-2 **804** may include a new data indicator **828**. The SCI-2 **804** may include an indication of a redundancy version **830**. The SCI-2 **804** may include a source ID **832** corresponding to a UE that is the source of the SCI-2 **804** (i.e., a UE that transmitted the SCI-2 **804**). The SCI-2 **804** may include a destination ID **834** corresponding to a UE (or a group of UEs) that is/are the intended destination of the SCI-2 **804** (i.e., a UE that is to receive the SCI-2 **804**). The SCI-2 **804** may include a channel state information (CSI) request **836**.

[0095] As noted above, a UE may utilize a C-DRX cycle for sidelink communications in order to conserve power. When a physical layer is indicated within an active time of an Rx UE from a medium access control (MAC) layer for candidate resource selection, a restriction may be applied in a physical layer such that at least a subset of candidate resources reported to the MAC layer may be located within an indicated active time of the Rx UE. If the candidate resources are not within an active time of the Rx UE, the Rx UE may add at least one resource within the active time (e.g., based on an implementation of the Rx UE). In unlicensed band sidelink (SL-U), a UE may perform a LBT procedure before transmission. In an example, the UE may first select a set of resources (e.g., time and frequency resources) and the UE may then fail the LBT procedure, which may trigger a LBT failure at a MAC layer of the UE. The UE may handle the LBT failure in different manners. In a first example, the UE may handle the LBT failure by issuing a resource reselection at the MAC layer. In a second example, the UE may handle the LBT failure by using a retransmission occasion for an initial transmission. If C-DRX is used in SL-U, resources selected by a UE implementation may be affected by interference. As a result, a probability of a failure of a LBT procedure may be higher. Furthermore, frequent resource reselection and/or insufficient resources may impede throughput at the UE.

[0096] Various technologies pertaining to enhancing a sidelink DRX cycle are described herein. In an example, a first UE obtains a configuration associated with a DRX cycle of a second UE, where the configuration includes a first

indication of a first set of resources (e.g., resources associated with a DRX on-duration of the second UE) and at least one condition, where the first set of resources is different from a second set of resources (e.g., resources associated with an extension of the DRX on-duration) utilized for the DRX cycle of the second UE. The first UE transmits, to the second UE based on an occurrence of the at least one condition, SL data via the first set of resources or the second set of resources. Vis-à-vis the configuration, the above-described technologies may increase communications reliability between the first UE and the second UE. For instance, the configuration may reduce occurrences of resource reselection at a MAC layer and/or may reduce use of retransmission occasions for initial transmissions. The configuration may also reduce a probability of a LBT failure via reduced resource reselection.

[0097] FIG. 9 is a diagram 900 illustrating an example of a DRX on-duration and an extended DRX on-duration. The diagram 900 depicts a DRX cycle n , a DRX cycle $n+1$, and a DRX cycle $n+2$, where n is an integer. A Tx UE may select a set of additional candidate resources within an extended DRX on-duration as an alternate in a resource selection procedure. The DRX on-duration extension may be activated when an implicit indication is met in the DRX on-duration. If the Tx UE fails the LBT procedure or the Tx UE does not have sufficient resources for transmissions, the DRX on-duration extension may be activated and the Tx UE may use the additional candidate resources to communicate with a Rx UE without utilizing a resource reselection. If the LBT procedure fails before the selected resource in the extended DRX on-duration, the Tx UE may trigger resource reselection.

[0098] As illustrated in the diagram 900, at 906, a Rx UE (e.g., the Rx UE 506) may detect a SCI within a DRX on-duration 902. In an example, the DRX on-duration 902 may correspond to the long DRX on-duration 402 described above. As a result, an extended DRX on-duration 904 may not be activated. Stated differently, the extended DRX on-duration 904 may not be activated as an implicit condition is not met.

[0099] At 908, the Rx UE may not detect a SCI during the DRX on-duration 902. As a result, at 910, the extended DRX on-duration 904 may be activated. Stated differently, the extended DRX on-duration 904 may be activated as an implicit condition is met (e.g., failing to detect the SCI). The Rx UE and a Tx UE (e.g., the Tx UE 504) may communicate via resources associated with the extended DRX on-duration 904 without utilizing resource reselection.

[0100] In an example, at 912, the Rx UE may not detect a SCI during the DRX on-duration 902. As a result, at 914, the extended DRX on-duration 904 may be activated. Stated differently, the extended DRX on-duration 904 may be activated as an implicit condition is met (e.g., failing to detect the SCI). In the example, at 916, during the extended DRX on-duration 904, a LBT procedure may fail. In the example, at 918, the Tx UE may trigger resource reselection based on the failure of the LBT procedure.

[0101] FIG. 10 is a diagram 1000 illustrating examples of an offset 1002, a duration 1004, a cycle length 1006, and a cycle timer 1008 associated with the extended DRX on-duration 904. The offset 1002 may refer to a time difference between a starting point of the extended DRX on-duration 904 and a starting point of the DRX on-duration 902. The duration 1004 may refer to an active time of the extended

DRX on-duration 904. The cycle length 1006 may refer to a period of the extended DRX on-duration 904. The cycle timer 1008 may refer to a number of DRX-on duration extensions between two DRX on-durations. In an example, the cycle timer 1008 may be equal to three. Multiple DRX on-duration extension may be associated with LBT diversity gain which may mitigate LBT failures.

[0102] FIG. 11 is a diagram 1100 illustrating an example of a fixed length extended DRX on-duration. The extended DRX on-duration 904 may be configured via a PC-5 RRC configuration message. In an example, the PC-5 RRC configuration message may indicate the offset 1002, the duration 1004, the cycle length 1006, and/or the cycle timer 1008 described above. In an example, a Tx UE (e.g., the Tx UE 504) may determine the extended DRX on-duration 904 for an associated Rx UE (e.g., the Rx UE 506).

[0103] A time period (i.e., a duration) of the extended DRX on-duration 904 may be fixed. In an example, the time period of the extended DRX on-duration 904 may be configured directly via a parameter in a PC-5 RRC configuration message. When an implicit indication (e.g., a condition) is met within the DRX on-duration 902, the Rx UE may extend the DRX on-duration immediately with a value from the PC-5 RRC configuration message.

[0104] As illustrated in the diagram 1100, an implicit indication may occur at 1102. Stated differently, a condition may occur at 1102. In response to the implicit indication, a UE (e.g., the Rx UE 506) may activate the extended DRX on-duration 904. The extended DRX on-duration 904 may be based on a PC-5 RRC configured value 1104.

[0105] FIG. 12 is a diagram 1200 illustrating an example of a dynamic length extended DRX on-duration. A time period (i.e., a duration) of the extended DRX on-duration 904 may be dynamically changed. In an example, the time period of the total active time may be configured via a parameter in a PC-5 RRC configuration message. When an implicit indication is met (e.g., when a condition occurs), in the DRX on-duration 902, a Rx UE (e.g., the Rx UE 506) may extend the DRX on-duration 902 according to equation (I) below.

$$T_{DRX_Extension} = T_{Configured_DRX} - T_{Active_in_DRX_on_duration} \quad (I)$$

[0106] In equation (I), $T_{DRX_Extension}$ may refer to a time period (i.e., a duration) of the extended DRX on-duration 904, $T_{Configured_DRX}$ may refer to a time period (i.e., a duration) of the DRX on-duration that is configured via a PC-5 RRC message, and $T_{Active_in_DRX_on_duration}$ may refer to an amount of time spent in the DRX on-duration 902 when the implicit indication occurs. A total active time in the DRX on-duration 902 and the extended DRX on-duration 904 may be equal to the configured DRX on-duration.

[0107] As illustrated in the diagram 1200, an implicit indication may occur at 1202 within the DRX on-duration 902. Stated differently, a condition may occur at 1202. In response to the implicit indication, a UE (e.g., the Rx UE 506) may extend the DRX on-duration according to equation (I) above based on a PC-5 RRC configured value 1204.

[0108] FIG. 13 is a diagram 1300 illustrating examples of extended DRX on-duration implicit indications 1302. The extended DRX on-duration implicit indications 1302 may be referred to as “conditions” or as “implicit conditions.” The

extended DRX on-duration implicit indications **1302** may include a first implicit indication **1304**, a second implicit indication **1306**, a third implicit indication **1308**, a fourth implicit indication **1310**, and a fifth implicit indication **1312**. When an indication in the extended DRX on-duration implicit indications **1302** occurs/is met, a UE may extend a DRX on-duration, that is, the UE may activate an extended DRX on-duration (e.g., the extended DRX on-duration **904**).

[0109] In the first implicit indication **1304**, a Rx UE (e.g., the Rx UE **506**) may fail to detect a SCI-2 (e.g., the SCI-2 **804**) during a DRX on-duration (e.g., the DRX on-duration **902**), where a source ID (e.g., the source ID **832**) of the SCI-2 matches a Tx UE (e.g., the Tx UE **504**). When the first implicit indication **1304** occurs/is met, the Rx UE may extend a DRX on-duration after an ending point of the DRX on-duration. In an example, the Rx UE may extend the DRX on-duration as described above in the description of FIG. **11** or FIG. **12**.

[0110] In the second implicit indication **1306**, a Rx UE (e.g., the Rx UE **506**) may fail to detect a SCI-2 (e.g., the SCI-2 **804**) during a DRX on-duration (e.g., the DRX on-duration **902**), where a source ID (e.g., the source ID **832**) of the SCI-2 matches a Tx UE (e.g., the Tx UE **504**) and a destination ID (e.g., the destination ID **834**) matches the Rx UE. When the second implicit indication **1306** occurs/is met, the Rx UE may extend a DRX on-duration after an ending point of the DRX on-duration. In an example, the Rx UE may extend the DRX on-duration as described above in the description of FIG. **11** or FIG. **12**.

[0111] In the third implicit indication **1308**, a Rx UE (e.g., the Rx UE **506**) may fail to detect a SCI-1 (e.g., the SCI-1 **802**) during a DRX on-duration (e.g., the DRX on-duration **902**). When the third implicit indication **1308** occurs/is met, the Rx UE may extend a DRX on-duration after an ending point of the DRX on-duration. In an example, the Rx UE may extend the DRX on-duration as described above in the description of FIG. **11** or FIG. **12**.

[0112] The third implicit indication **1308** may be associated with less UE wake up in comparison to the first implicit indication **1304** and/or the second implicit indication **1306**. The third implicit indication **1308** may be associated with less radio wake up compared with the first implicit indication **1304** and the second implicit indication **1306**. The third implicit indication **1308** may be useful for inter-radio access technology (inter-RAT) contention. In SL-U, if a Tx UE is able to perform a transmission, the Tx UE may send a SCI-1. However, if a Wi-Fi radio acquires a channel, the Tx UE may not transmit a SL-U transmission and the Tx UE may not send a SCI-1 and a Rx UE may extend a DRX on-duration. If an extended DRX on-duration is longer than a transmission ending point associated with a Wi-Fi transmission, SL-U devices transmitting in SL-U can perform transmissions and hence reduce latency. If a SCI-1 is detected by a SL-U UE, at least one SL-U UE (e.g., UE A) may be transmitting and there may be no Wi-Fi transmission. In an example, if there is relaxed COT sharing, UE A can share the COT with an intended transmitter (e.g., UE B) such that UE B can perform a transmission to the Rx UE in the DRX on-duration without extending the DRX on-duration.

[0113] In the fourth implicit indication **1310**, a Rx UE (e.g., the Rx UE **506**) may detect a SCI-2 (e.g., the SCI-2 **804**) during a DRX on-duration (e.g., the DRX on-duration **902**), where a source ID (e.g., the source ID **832**) of the SCI-2 matches a Tx UE (e.g., the Tx UE **504**). When the

fourth implicit indication **1310** occurs/is met, the Rx UE may activate a DRX on-duration extension (e.g., the extended DRX on-duration **904**). In an example, the Rx UE may extend the DRX on-duration as described above in the description of FIG. **11** or FIG. **12**. The Rx UE may monitor for an SCI (e.g., a second SCI-2) until the end of the extended DRX on-duration.

[0114] In the fifth implicit indication **1312**, a Rx UE (e.g., the Rx UE **506**) may detect a SCI-2 (e.g., the SCI-2 **804**) during a DRX on-duration (e.g., the DRX on-duration **902**), where a source ID (e.g., the source ID **832**) of the SCI-2 matches a Tx UE (e.g., the Tx UE **504**) and a destination ID (e.g., the destination ID **834**) matches the Rx UE. When the fifth implicit indication **1312** occurs/is met, the Rx UE may activate a DRX on-duration extension (e.g., the extended DRX on-duration **904**). In an example, the Rx UE may extend the DRX on-duration as described above in the description of FIG. **11** or FIG. **12**. The Rx UE may monitor for an SCI (e.g., a second SCI-2) until the end of the extended DRX on-duration.

[0115] FIG. **14** is a diagram **1400** illustrating an example of utilizing a fixed length extended DRX on-duration upon detection of SCIs. Aspects of the diagram **1400** may correspond to the fourth implicit indication **1310** or the fifth implicit indication **1312** described above. For a fixed length extended DRX on-duration (e.g., such as the fixed length extended DRX on-duration described above in the description of FIG. **11**), a Rx UE (e.g., the Rx UE **506**) may activate a first extended DRX on-duration **1408** immediately after detecting a first SCI-2 at **1402**. If the Rx UE detects a second SCI-2 at **1404** (either during the DRX on-duration **902** or during the first extended DRX on-duration **1408**), the Rx UE may activate a second extended DRX on-duration **1410**. The first extended DRX on-duration **1408** and the second extended DRX on-duration **1410** may be based on a PC-5 RRC configured value **1406**.

[0116] FIG. **15** is a diagram **1500** illustrating an example of utilizing a dynamic length extended DRX on-duration upon detection of SCIs. Aspects of the diagram **1500** may correspond to the fourth implicit indication **1310** or the fifth implicit indication **1312** described above. For a dynamic length extended DRX on-duration (e.g., such as the dynamic length extended DRX on-duration described above in the description of FIG. **12**), a Rx UE (e.g., the Rx UE **506**) may activate the extended DRX on-duration **904** at an end of the DRX on-duration **902**. In the example depicted in the diagram **1500**, the Rx UE may receive a first SCI-2 at **1502** and a second SCI-2 at **1504**. An active time in the DRX on-duration may be determined based on the first SCI-2 (i.e., the first detected SCI-2). The extended DRX on-duration **904** may be based on a PC-5 RRC configured value **1506**.

[0117] FIG. **16** is a diagram **1600** illustrating example communications between a first UE **1602** and a second UE **1604**. In an example, the first UE **1602** may be the Tx UE **504** and the second UE **1604** may be the Rx UE **506**.

[0118] At **1608**, the first UE **1602** may obtain a configuration associated with a DRX cycle of the second UE **1604**. The configuration may indicate first resources and second resources (e.g., time and frequency resources). The configuration may also indicate a condition (i.e., an implicit indication). The condition may be or include one or more of the extended DRX on-duration implicit indications **1302**. In an example, the first resources may be associated with a DRX on-duration (e.g., the DRX on-duration **902**) and the second

resources may be associated with an extension of the DRX on-duration (e.g., the extended DRX on-duration 904). In one aspect, at 1610, the first UE 1602 may transmit the configuration to the second UE 1604.

[0119] At 1612, the first UE 1602 may transmit an indication of a condition (e.g., an implicit indication) via SCI or a PC-5 RRC message. In one example, if the condition corresponds to the first implicit indication 1304, the second implicit indication 1306, the fourth implicit indication 1310, or the fifth implicit indication 1312, the first UE 1602 may transmit the indication of the condition via a SCI-2. In another example, if the condition corresponds to the first implicit indication 1304, the second implicit indication 1306, the third implicit indication 1308, the fourth implicit indication 1310, or the fifth implicit indication 1312, the first UE 1602 may transmit the indication of the condition via a PC-5 RRC configuration message. At 1614, the second UE 1604 may monitor for SL data from the first UE 1602 via the first resources. The second UE 1604 may monitor for the SL data based on the configuration. In an example, monitoring for the SL data via the first resources may include monitoring for the SL data during a DRX on-duration (e.g., the DRX on-duration 902).

[0120] At 1616, the second UE detects an occurrence of a condition. The conditions may be or include one or more of the extended DRX on-duration implicit indications 1302. In a first example, at 1616A, the second UE 1604 may fail to detect a SCI-2 (e.g., the SCI-2 804). For instance, the second UE 1604 may fail to detect the SCI-2 during a DRX on-duration (e.g., the DRX on-duration 902). In a second example, at 1616B, the second UE 1604 may fail to detect a SCI-1 (e.g., the SCI-1 802). For instance, the second UE 1604 may fail to detect the SCI-1 during a DRX on-duration (e.g., the DRX on-duration 902). In a third example, at 1616C, the second UE 1604 may detect a SCI-2 (e.g., the SCI-2 804). For instance, the second UE 1604 may detect the SCI-2 during a DRX on-duration (e.g., the DRX on-duration 902). In one aspect, at 1618, the second UE 1604 may determine a time period for an extension of the DRX on-duration (e.g., the extended DRX on-duration 904). In an example, the second UE 1604 may determine the time period for the extension of the DRX on-duration dynamically as described above in the description of FIG. 12). In one aspect, at 1620, the second UE 1604 may use a time period for the extension of the DRX on-duration, where the time period may be included in the configuration. In an example, the second UE 1604 may utilize a fixed length extension of the DRX on-duration as described above in the description of FIG. 11.

[0121] At 1622, the second UE 1604 may monitor for SL data via the second resources. The second resources may be associated with an extension of the DRX on-duration (e.g., the extended DRX on-duration 904). The second resources may be associated with a time period determined at 1618 or a time period used from the configuration at 1620. At 1624, the first UE 1602 may transmit SL data via the first resources or the second resources. In an example, the SL data may include SL control signaling.

[0122] FIG. 17 is a flowchart 1700 of a method of wireless communication. The method may be performed by a first UE (e.g., the UE 104, the UE 350, the Tx UE 504, the first UE 1602, the apparatus 2104). The method may be associated with various advantages at the first UE, such as increased communications reliability with a second UE. For instance,

the method may be associated with reduced resource reselections by the first UE. In an example, the method may be performed by the DRX component 198.

[0123] At 1702, the first UE obtains a configuration associated with a DRX cycle of a second UE, where the configuration includes a first indication of a first set of resources and at least one condition, where the first set of resources is different from a second set of resources utilized for the DRX cycle of the second UE. For example, FIG. 16 at 1608 shows that the first UE 1602 may obtain a configuration associated with a DRX cycle of the second UE 1604, where the configuration indicates first resources and second resources. In an example, the DRX cycle may be the (long) DRX cycle 404 illustrated in FIG. 4. In yet another example, the DRX cycle may be associated with the DRX cycles illustrated in FIG. 9. In an example, the at least one condition may be one or more of the extended DRX on-duration implicit indications 1302. In an example, 1702 may be performed by the DRX component 198.

[0124] At 1704, the first UE transmits, to the second UE based on an occurrence of the at least one condition, SL data via the first set of resources or the second set of resources. For example, FIG. 16 at 1624 shows that the first UE 1602 may transmit SL data to the second UE 1604 via first resources or second resources. In an example, 1704 may be performed by the DRX component 198.

[0125] FIG. 18 is a flowchart 1800 of a method of wireless communication. The method may be performed by a first UE (e.g., the UE 104, the UE 350, the Tx UE 504, the first UE 1602, the apparatus 2104). The method may be associated with various advantages at the first UE, such as increased communications reliability with a second UE. For instance, the method may be associated with reduced resource reselections by the first UE. In an example, the method (including the various aspects detailed below) may be performed by the DRX component 198.

[0126] At 1802, the first UE obtains a configuration associated with a DRX cycle of a second UE, where the configuration includes a first indication of a first set of resources and at least one condition, where the first set of resources is different from a second set of resources utilized for the DRX cycle of the second UE. For example, FIG. 16 at 1608 shows that the first UE 1602 may obtain a configuration associated with a DRX cycle of the second UE 1604, where the configuration indicates first resources and second resources. In an example, the DRX cycle may be the (long) DRX cycle 404 illustrated in FIG. 4. In yet another example, the DRX cycle may be associated with the DRX cycles illustrated in FIG. 9. In an example, the condition may be one or more of the extended DRX on-duration implicit indications 1302. In an example, 1802 may be performed by the DRX component 198.

[0127] At 1806, the first UE transmits, to the second UE based on an occurrence of the at least one condition, SL data via the first set of resources or the second set of resources. For example, FIG. 16 at 1624 shows that the first UE 1602 may transmit SL data to the second UE 1604 via first resources or second resources. In an example, 1806 may be performed by the DRX component 198.

[0128] In one aspect, the first set of resources may be associated with a DRX on-duration of the second UE, where the second set of resources may be associated with an extension of the DRX on-duration. For example, the first set of resources may be associated with the DRX on-duration

902 and the second set of resources may be associated with the extended DRX on-duration **904**.

[0129] In one aspect, the extension of the DRX on-duration may be triggered via a second indication. For example, FIG. 16 at **1616A** shows that an extension of a DRX on-duration may be triggered by a SCI-2 not being detected. In another example, FIG. 16 at **1616B** shows that an extension of a DRX on-duration may be triggered by a SCI-1 not being detected. In yet another example, FIG. 16 at **1616C** shows that an extension of a DRX on-duration may be triggered by a SCI-2 being detected.

[0130] In one aspect, the second set of resources may be activated based on the second indication. For example, resources corresponding to the extended DRX on-duration **904** may be activated based on a second indication.

[0131] In one aspect, a time period of the extension of the DRX on-duration may be included in the configuration or the time period of the extension of the DRX on-duration may be based on a difference between a first time period of a configured DRX on-duration and a second time period corresponding to an active time in the DRX on-duration. For example, the configuration obtained by the first UE **1602** may include a time period of an extension of a DRX on-duration. In another example, FIG. 11 illustrates that a time period of the extension of the DRX on-duration **904** may be fixed. In yet another example, FIG. 12 illustrates that a time period of the extension of the DRX on-duration **904** may be dynamic and based on an implicit indication occurring at **1202** and a PC-5 RRC configured value **1204**.

[0132] In one aspect, the at least one condition may include: a lack of a detection of SCI-1 during the DRX on-duration by the second UE, a lack of a detection of SCI-2 during the DRX on-duration by the second UE, or a detection of the SCI-2 during the DRX on-duration by the second UE. For example, FIG. 16 at **1616A** shows that a condition may include a SCI-2 not being detected. In another example, FIG. 16 at **1616B** shows that a condition may include a SCI-1 not being detected. In yet another example, FIG. 16 at **1616C** shows that a condition may include a SCI-2 being detected. In another example, the SCI-1 may be the SCI-1 **802** and the SCI-2 may be the SCI-2 **804**. In a further example, the at least one condition may be one or more of the first implicit indication **1304**, the second implicit indication **1306**, the third implicit indication **1308**, the fourth implicit indication **1310**, or the fifth implicit indication **1312**.

[0133] In one aspect, the SCI-2 may include a second indication of a source ID that corresponds to the first UE. For example, FIG. 13 shows that in the first implicit indication **1304**, a source ID of a SCI-2 may match a Tx UE (e.g., the first UE **1602**). In another example, FIG. 13 shows that in the fourth implicit indication **1310**, a source ID of the SCI-2 may match a Tx UE. In another example, the source ID may be the source ID **832** of the SCI-2 **804**.

[0134] In one aspect, the SCI-2 may include a second indication of a source ID that corresponds to the first UE and a destination ID that corresponds to the second UE. For example, FIG. 13 shows that in the second implicit indication **1306**, a source ID of a SCI-2 may match a Tx UE (e.g., the first UE **1602**) and a destination ID of the SCI-2 may match a Rx UE (e.g., the second UE **1604**). In another example, FIG. 13 shows that in the fifth implicit indication **1312**, a source ID of the SCI-2 may match a Tx UE and a destination ID of the SCI-2 may match a Rx UE. In another

example, the source ID may be the source ID **832** of the SCI-2 **804** and the destination ID may be the destination ID **834**.

[0135] In one aspect, the extension of the DRX on-duration may be activated immediately upon the second UE detecting the SCI-2 during the DRX on-duration, or the extension of the DRX on-duration may be activated upon the second UE detecting the SCI-2 during the DRX on-duration and after the DRX on-duration has expired. For example, FIG. 14 shows that the first extended DRX on-duration **1408** and the second extended DRX on-duration **1410** may be activated immediately upon detection of SCI-2 at **1402** and **1404**. In another example, FIG. 15 shows that the extended DRX on-duration **904** may be activated upon a second UE detecting an SCI-2 at **1502** and after the DRX on-duration **902** has expired.

[0136] In one aspect, at **1803**, the first UE may transmit, to the second UE, the configuration associated with the DRX cycle prior to transmitting the SL data. For example, FIG. 16 at **1610** shows that the first UE **1602** may transmit the configuration associated with the DRX cycle of the second UE **1604** prior to the SL data being transmitted at **1624**. In an example, **1803** may be performed by the DRX component **198**.

[0137] In one aspect, the SL data may be SL control signaling. For example, FIG. 16 at **1624** shows that the SL data may be SL control signaling.

[0138] In one aspect, at **1804**, the first UE may transmit, to the second UE, a second indication of the at least one condition via SCI or PC-5 RRC signaling. For example, FIG. 16 at **1612** shows that the first UE **1602** may transmit an indication of a condition to the second UE **1604** via a SCI or PC-5 RRC signaling. In an example, **1804** may be performed by the DRX component **198**.

[0139] FIG. 19 is a flowchart **1900** of a method of wireless communication. The method may be performed by a second UE (e.g., the UE **104**, the UE **350**, the Rx UE **506**, the second UE **1604**, the apparatus **2104**). The method may be associated with various advantages at the second UE, such as increased communications reliability with a first UE. In an example, the method may be performed by the DRX component **198**.

[0140] At **1902**, the second UE monitors for SL data via a first set of resources, where the first set of resources is different from a second set of resources utilized for a DRX cycle of the second UE. For example, FIG. 16 at **1614** shows that the second UE **1604** may monitor for SL data via first resources. In an example, the DRX cycle may be the (long) DRX cycle **404** illustrated in FIG. 4. In yet another example, the DRX cycle may be associated with the DRX cycles illustrated in FIG. 9. In an example, **1902** may be performed by the DRX component **198**.

[0141] At **1904**, the second UE receives, from a first UE based on an occurrence of at least one condition, the SL data via the first set of resources or the second set of resources. For example, FIG. 16 at **1624** shows that the second UE **1604** may receive SL data from the first UE **1602** via first resources or second resources. In an example, the at least one condition may be one or more of the extended DRX on-duration implicit indications **1302**. In an example, **1904** may be performed by the DRX component **198**.

[0142] FIG. 20 is a flowchart **2000** of a method of wireless communication. The method may be performed by a second UE (e.g., the UE **104**, the UE **350**, the Rx UE **506**, the

second UE **1604**, the apparatus **2104**). The method may be associated with various advantages at the second UE, such as increased communications reliability with a first UE. In an example, the method (including the various aspects detailed below) may be performed by the DRX component **198**.

[0143] At **2006**, the second UE monitors for SL data via a first set of resources, where the first set of resources is different from a second set of resources utilized for a DRX cycle of the second UE. For example, FIG. **16** at **1614** shows that the second UE **1604** may monitor for SL data via first resources. In an example, the DRX cycle may be the (long) DRX cycle **404** illustrated in FIG. **4**. In yet another example, the DRX cycle may be associated with the DRX cycles illustrated in FIG. **9**. In an example, **2006** may be performed by the DRX component **198**.

[0144] At **2014**, the second UE receives, from a first UE based on an occurrence of at least one condition, the SL data via the first set of resources or the second set of resources. For example, FIG. **16** at **1624** shows that the second UE **1604** may receive SL data from the first UE **1602** via first resources or second resources. In an example, the at least one condition may be one or more of the extended DRX on-duration implicit indications **1302**. In an example, **2014** may be performed by the DRX component **198**.

[0145] In one aspect, the first set of resources may be associated with a DRX on-duration of the second UE, where the second set of resources may be associated with an extension of the DRX on-duration. For example, the first set of resources may be associated with the DRX on-duration **902** and the second set of resources may be associated with the extended DRX on-duration **904**.

[0146] In one aspect, the extension of the DRX on-duration may be triggered via a second indication. For example, FIG. **16** at **1616A** shows that an extension of a DRX on-duration may be triggered by a SCI-2 not being detected. In another example, FIG. **16** at **1616B** shows that an extension of a DRX on-duration may be triggered by a SCI-1 not being detected. In yet another example, FIG. **16** at **1616C** shows that an extension of a DRX on-duration may be triggered by a SCI-2 being detected.

[0147] In one aspect, the second set of resources may be activated based on the second indication. For example, resources corresponding to the extended DRX on-duration **904** may be activated based on a second indication.

[0148] In one aspect, a time period of the extension of the DRX on-duration may be included in a configuration associated with the DRX cycle of the second UE. For example, the configuration transmitted by the second UE **1604** may include a time period of an extension of a DRX on-duration. In another example, FIG. **11** illustrates that a time period of the extension of the DRX on-duration **904** may be fixed.

[0149] In one aspect, at **2008**, the second UE may determine a time period of the extension of the DRX on-duration based on a difference between a first time period of a configured DRX on-duration and a second time period corresponding to an active time in the DRX on-duration. For example, FIG. **16** at **1620** shows that the second UE **1604** may determine a time period for an extension of a DRX on-duration (e.g., based on a difference between a first time period of a configured DRX on-duration and a second time period corresponding to an active time in the DRX on-duration). In another example, FIG. **12** illustrates that a time period of the extension of the DRX on-duration **904** may be

dynamic and based on an implicit indication occurring at **1202** and a PC-5 RRC configured value **1204**. In an example, **2008** may be performed by the DRX component **198**.

[0150] In one aspect, at **2010**, the second UE may detect the occurrence of the at least one condition. For example, FIG. **16** at **1616** shows that the second UE **1604** may detect an occurrence of a condition. In an example, **2010** may be performed by the DRX component **198**.

[0151] In one aspect, at **2012**, the second UE may monitor for the SL data via the second set of resources based upon the detection of the occurrence of the at least one condition. For example, FIG. **16** at **1622** shows that the second UE **1604** may monitor for SL data via second resources upon detecting the occurrence of the condition at **1616**. In an example, **2012** may be performed by the DRX component **198**.

[0152] In one aspect, the at least one condition may include: a lack of a detection of SCI-1 during the DRX on-duration by the second UE, a lack of a detection of SCI-2 during the DRX on-duration by the second UE, or a detection of the SCI-2 during the DRX on-duration by the second UE. For example, FIG. **16** at **1616A** shows that a condition may include a SCI-2 not being detected. In another example, FIG. **16** at **1616B** shows that a condition may include a SCI-1 not being detected. In yet another example, FIG. **16** at **1616C** shows that a condition may include a SCI-2 being detected. In another example, the SCI-1 may be the SCI-1 **802** and the SCI-2 may be the SCI-2 **804**. In a further example, the at least one condition may be one or more of the first implicit indication **1304**, the second implicit indication **1306**, the third implicit indication **1308**, the fourth implicit indication **1310**, or the fifth implicit indication **1312**.

[0153] In one aspect, the SCI-2 may include a second indication of a source ID that corresponds to the first UE. For example, FIG. **13** shows that in the first implicit indication **1304**, a source ID of a SCI-2 may match a Tx UE (e.g., the first UE **1602**). In another example, FIG. **13** shows that in the fourth implicit indication **1310**, a source ID of the SCI-2 may match a Tx UE. In another example, the source ID may be the source ID **832** of the SCI-2 **804**.

[0154] In one aspect, the SCI-2 may include a second indication of a source ID that corresponds to the first UE and a destination ID that corresponds to the second UE. For example, FIG. **13** shows that in the second implicit indication **1306**, a source ID of a SCI-2 may match a Tx UE (e.g., the first UE **1602**) and a destination ID of the SCI-2 may match a Rx UE (e.g., the second UE **1604**). In another example, FIG. **13** shows that in the fifth implicit indication **1312**, a source ID of the SCI-2 may match a Tx UE and a destination ID of the SCI-2 may match a Rx UE. In another example, the source ID may be the source ID **832** of the SCI-2 **804** and the destination ID may be the destination ID **834**.

[0155] In one aspect, the extension of the DRX on-duration may be activated immediately upon the second UE detecting the SCI-2 during the DRX on-duration, or the extension of the DRX on-duration may be activated upon the second UE detecting the SCI-2 during the DRX on-duration and after the DRX on-duration has expired. For example, FIG. **14** shows that the first extended DRX on-duration **1408** and the second extended DRX on-duration **1410** may be activated immediately upon detection of SCI-2 at **1402** and

1404. In another example, FIG. 15 shows that the extended DRX on-duration **904** may be activated upon a second UE detecting an SCI-2 at **1502** and after the DRX on-duration **902** has expired.

[0156] In one aspect, at **2002**, the second UE may receive, from the first UE, a configuration associated with the DRX cycle prior to receiving the SL data. For example, FIG. 16 at **1610** shows that the second UE **1604** may receive a configuration associated with a DRX cycle prior to receiving the SL data at **1624**. In an example, **2002** may be performed by the DRX component **198**.

[0157] In one aspect, the SL data may be SL control signaling. For example, FIG. 16 at **1624** shows that the SL data may be SL control signaling.

[0158] In one aspect, at **2004**, the second UE may receive a second indication of the at least one condition via SCI or PC-5 RRC signaling. For example, FIG. 16 at **1612** shows that the second UE **1604** may receive an indication of a condition from the first UE **1602** via a SCI or PC-5 RRC signaling. In an example, **2004** may be performed by the DRX component **198**.

[0159] FIG. 21 is a diagram **2100** illustrating an example of a hardware implementation for an apparatus **2104**. The apparatus **2104** may be a UE, a component of a UE, or may implement UE functionality. In some aspects, the apparatus **2104** may include a cellular baseband processor **2124** (also referred to as a modem) coupled to one or more transceivers **2122** (e.g., cellular RF transceiver). The cellular baseband processor **2124** may include on-chip memory **2124'**. In some aspects, the apparatus **2104** may further include one or more subscriber identity modules (SIM) cards **2120** and an application processor **2106** coupled to a secure digital (SD) card **2108** and a screen **2110**. The application processor **2106** may include on-chip memory **2106'**. In some aspects, the apparatus **2104** may further include a Bluetooth module **2112**, a WLAN module **2114**, an SPS module **2116** (e.g., GNSS module), one or more sensor modules **2118** (e.g., barometric pressure sensor/altimeter; motion sensor such as inertial measurement unit (IMU), gyroscope, and/or accelerometer (s); light detection and ranging (LIDAR), radio assisted detection and ranging (RADAR), sound navigation and ranging (SONAR), magnetometer, audio and/or other technologies used for positioning), additional memory modules **2126**, a power supply **2130**, and/or a camera **2132**. The Bluetooth module **2112**, the WLAN module **2114**, and the SPS module **2116** may include an on-chip transceiver (TRX) (or in some cases, just a receiver (RX)). The Bluetooth module **2112**, the WLAN module **2114**, and the SPS module **2116** may include their own dedicated antennas and/or utilize the antennas **2180** for communication. The cellular baseband processor **2124** communicates through the transceiver(s) **2122** via one or more antennas **2180** with the UE **104** and/or with an RU associated with a network entity **2102**. The cellular baseband processor **2124** and the application processor **2106** may each include a computer-readable medium/memory **2124'**, **2106'**, respectively. The additional memory modules **2126** may also be considered a computer-readable medium/memory. Each computer-readable medium/memory **2124'**, **2106'**, **2126** may be non-transitory. The cellular baseband processor **2124** and the application processor **2106** are each responsible for general processing, including the execution of software stored on the computer-readable medium/memory. The software, when executed by the cellular baseband processor **2124**/applica-

tion processor **2106**, causes the cellular baseband processor **2124**/application processor **2106** to perform the various functions described supra. The computer-readable medium/memory may also be used for storing data that is manipulated by the cellular baseband processor **2124**/application processor **2106** when executing software. The cellular baseband processor **2124**/application processor **2106** may be a component of the UE **350** and may include the memory **360** and/or at least one of the TX processor **368**, the RX processor **356**, and the controller/processor **359**. In one configuration, the apparatus **2104** may be a processor chip (modem and/or application) and include just the cellular baseband processor **2124** and/or the application processor **2106**, and in another configuration, the apparatus **2104** may be the entire UE (e.g., see **350** of FIG. 3) and include the additional modules of the apparatus **2104**.

[0160] As discussed supra, the DRX component **198** is configured to obtain a configuration associated with a DRX cycle of a second UE, where the configuration includes a first indication of a first set of resources and at least one condition, where the first set of resources is different from a second set of resources utilized for the DRX cycle of the second UE. The DRX component **198** is configured to transmit, to the second UE based on an occurrence of the at least one condition, SL data via the first set of resources or the second set of resources. The DRX component **198** is configured to transmit, to the second UE, the configuration associated with the DRX cycle prior to transmit the SL data. The DRX component **198** is configured to transmit, to the second UE, a second indication of the at least one condition via SCI or PC-5 RRC signaling. The DRX component **198** is configured to monitor for SL data via a first set of resources, where the first set of resources is different from a second set of resources utilized for a DRX cycle of the second UE. The DRX component **198** is configured to receive, from a first UE based on an occurrence of at least one condition, the SL data via the first set of resources or the second set of resources. The DRX component **198** is configured to determine a time period of the extension of the DRX on-duration based on a difference between a first time period of a configured DRX on-duration and a second time period corresponding to an active time in the DRX on-duration. The DRX component **198** is configured to detect the occurrence of the at least one condition. The DRX component **198** is configured to monitor for the SL data via the second set of resources based upon the detection of the occurrence of the at least one condition. The DRX component **198** is configured to receive, from the first UE, a configuration associated with the DRX cycle prior to receive the SL data. The DRX component **198** is configured to receive a second indication of the at least one condition via SCI or PC-5 RRC signaling. The DRX component **198** may be within the cellular baseband processor **2124**, the application processor **2106**, or both the cellular baseband processor **2124** and the application processor **2106**. The DRX component **198** may be one or more hardware components specifically configured to carry out the stated processes/algorithm, implemented by one or more processors configured to perform the stated processes/algorithm, stored within a computer-readable medium for implementation by one or more processors, or some combination thereof. As shown, the apparatus **2104** may include a variety of components configured for various functions. In one configuration, the apparatus **2104**, and in particular the cellular baseband

processor **2124** and/or the application processor **2106**, includes means for obtaining a configuration associated with a DRX cycle of a second UE, where the configuration includes a first indication of a first set of resources and at least one condition, where the first set of resources is different from a second set of resources utilized for the DRX cycle of the second UE. In one configuration, the apparatus **2104**, and in particular the cellular baseband processor **2124** and/or the application processor **2106**, includes means for transmitting, to the second UE based on an occurrence of the at least one condition, SL data via the first set of resources or the second set of resources. In one configuration, the means for obtaining the configuration associated with the DRX cycle of the second UE include means for transmitting, to the second UE, the configuration associated with the DRX cycle prior to transmitting the SL data. In one configuration, the apparatus **2104**, and in particular the cellular baseband processor **2124** and/or the application processor **2106**, includes means for transmitting, to the second UE, a second indication of the at least one condition via SCI or PC-5 RRC signaling. In one configuration, the apparatus **2104**, and in particular the cellular baseband processor **2124** and/or the application processor **2106**, includes means for monitoring for SL data via a first set of resources, where the first set of resources is different from a second set of resources utilized for a DRX cycle of the second UE. In one configuration, the apparatus **2104**, and in particular the cellular baseband processor **2124** and/or the application processor **2106**, includes means for receiving, from a first UE based on an occurrence of at least one condition, the SL data via the first set of resources or the second set of resources. In one configuration, the apparatus **2104**, and in particular the cellular baseband processor **2124** and/or the application processor **2106**, includes means for determining a time period of the extension of the DRX on-duration based on a difference between a first time period of a configured DRX on-duration and a second time period corresponding to an active time in the DRX on-duration. In one configuration, the apparatus **2104**, and in particular the cellular baseband processor **2124** and/or the application processor **2106**, includes means for detecting the occurrence of the at least one condition. In one configuration, the apparatus **2104**, and in particular the cellular baseband processor **2124** and/or the application processor **2106**, includes means for monitoring for the SL data via the second set of resources based upon detecting the occurrence of the at least one condition. In one configuration, the apparatus **2104**, and in particular the cellular baseband processor **2124** and/or the application processor **2106**, includes means for receiving, from the first UE, a configuration associated with the DRX cycle prior to receiving the SL data. In one configuration, the apparatus **2104**, and in particular the cellular baseband processor **2124** and/or the application processor **2106**, includes means for receiving a second indication of the at least one condition via SCI or PC-5 RRC signaling. The means may be the DRX component **198** of the apparatus **2104** configured to perform the functions recited by the means. As described supra, the apparatus **2104** may include the TX processor **368**, the RX processor **356**, and the controller/processor **359**. As such, in one configuration, the means may be the TX processor **368**, the RX processor **356**, and/or the controller/processor **359** configured to perform the functions recited by the means.

[0161] FIG. 22 is a diagram **2200** illustrating an example of a hardware implementation for a network entity **2202**.

The network entity **2202** may be a BS, a component of a BS, or may implement BS functionality. The network entity **2202** may include at least one of a CU **2210**, a DU **2230**, or an RU **2240**. The network entity **2202** may include the CU **2210**; both the CU **2210** and the DU **2230**; each of the CU **2210**, the DU **2230**, and the RU **2240**; the DU **2230**; both the DU **2230** and the RU **2240**; or the RU **2240**. The CU **2210** may include a CU processor **2212**. The CU processor **2212** may include on-chip memory **2212'**. In some aspects, the CU **2210** may further include additional memory modules **2214** and a communications interface **2218**. The CU **2210** communicates with the DU **2230** through a midhaul link, such as an F1 interface. The DU **2230** may include a DU processor **2232**. The DU processor **2232** may include on-chip memory **2232'**. In some aspects, the DU **2230** may further include additional memory modules **2234** and a communications interface **2238**. The DU **2230** communicates with the RU **2240** through a fronthaul link. The RU **2240** may include an RU processor **2242**. The RU processor **2242** may include on-chip memory **2242'**. In some aspects, the RU **2240** may further include additional memory modules **2244**, one or more transceivers **2246**, antennas **2280**, and a communications interface **2248**. The RU **2240** communicates with the UE **104**. The on-chip memory **2212'**, **2232'**, **2242'** and the additional memory modules **2214**, **2234**, **2244** may each be considered a computer-readable medium/memory. Each computer-readable medium/memory may be non-transitory. Each of the processors **2212**, **2232**, **2242** is responsible for general processing, including the execution of software stored on the computer-readable medium/memory. The software, when executed by the corresponding processor(s) causes the processor(s) to perform the various functions described supra. The computer-readable medium/memory may also be used for storing data that is manipulated by the processor(s) when executing software.

[0162] As noted above, a UE may utilize a C-DRX cycle for sidelink communications in order to conserve power. When a physical layer is indicated within an active time of an Rx UE from a medium access control (MAC) layer for candidate resource selection, a restriction may be applied in a physical layer such that at least a subset of candidate resources reported to the MAC layer may be located within an indicated active time of the Rx UE. If the candidate resources are not within an active time of the Rx UE, the Rx UE may add at least one resource within the active time (e.g., based on an implementation of the Rx UE). In unlicensed band sidelink (SL-U), a UE may perform a LBT procedure before transmission. In an example, the UE may first select a set of resources (e.g., time and frequency resources) and the UE may then fail the LBT procedure, which may trigger a LBT failure at a MAC layer of the UE. The UE may handle the LBT failure in different manners. In a first example, the UE may handle the LBT failure by issuing a resource reselection at the MAC layer. In a second example, the UE may handle the LBT failure by using a retransmission occasion for an initial transmission. If C-DRX is used in SL-U, resources selected by a UE implementation may be affected by interference. As a result, a probability of a failure of a LBT procedure may be higher. Furthermore, frequent resource reselection and/or insufficient resources may impede throughput at the UE.

[0163] Various technologies pertaining to enhancing a sidelink DRX cycle are described herein. In an example, a first UE obtains a configuration associated with a DRX cycle

of a second UE, where the configuration includes a first indication of a first set of resources (e.g., resources associated with a DRX on-duration of the second UE) and at least one condition, where the first set of resources is different from a second set of resources (e.g., resources associated with an extension of the DRX on-duration) utilized for the DRX cycle of the second UE. The first UE transmits, to the second UE based on an occurrence of the at least one condition, SL data via the first set of resources or the second set of resources. Vis-à-vis the configuration, the above-described technologies may increase communications reliability between the first UE and the second UE. For instance, the configuration may reduce occurrences of resource reselection at a MAC layer and/or may reduce use of retransmission occasions for initial transmissions. The configuration may also reduce a probability of a LBT failure via reduced resource reselection.

[0164] It is understood that the specific order or hierarchy of blocks in the processes/flowcharts disclosed is an illustration of example approaches. Based upon design preferences, it is understood that the specific order or hierarchy of blocks in the processes/flowcharts may be rearranged. Further, some blocks may be combined or omitted. The accompanying method claims present elements of the various blocks in a sample order, and are not limited to the specific order or hierarchy presented.

[0165] The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not limited to the aspects described herein, but are to be accorded the full scope consistent with the language claims. Reference to an element in the singular does not mean “one and only one” unless specifically so stated, but rather “one or more.” Terms such as “if,” “when,” and “while” do not imply an immediate temporal relationship or reaction. That is, these phrases, e.g., “when,” do not imply an immediate action in response to or during the occurrence of an action, but simply imply that if a condition is met then an action will occur, but without requiring a specific or immediate time constraint for the action to occur. The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects. Unless specifically stated otherwise, the term “some” refers to one or more. Combinations such as “at least one of A, B, or C,” “one or more of A, B, or C,” “at least one of A, B, and C,” “one or more of A, B, and C,” “one or more of A, B, and C,” and “A, B, C, or any combination thereof” include any combination of A, B, and/or C, and may include multiples of A, multiples of B, or multiples of C. Specifically, combinations such as “at least one of A, B, or C,” “one or more of A, B, or C,” “at least one of A, B, and C,” “one or more of A, B, and C,” and “A, B, C, or any combination thereof” may be A only, B only, C only, A and B, A and C, B and C, or A and B and C, where any such combinations may contain one or more member or members of A, B, or C. Sets should be interpreted as a set of elements where the elements number one or more. Accordingly, for a set of X, X would include one or more elements. If a first apparatus receives data from or transmits data to a second apparatus, the data may be received/transmitted directly between the first and second apparatuses,

or indirectly between the first and second apparatuses through a set of apparatuses. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are encompassed by the claims. Moreover, nothing disclosed herein is dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. The words “module,” “mechanism,” “element,” “device,” and the like may not be a substitute for the word “means.” As such, no claim element is to be construed as a means plus function unless the element is expressly recited using the phrase “means for.” **[0166]** As used herein, the phrase “based on” shall not be construed as a reference to a closed set of information, one or more conditions, one or more factors, or the like. In other words, the phrase “based on A” (where “A” may be information, a condition, a factor, or the like) shall be construed as “based at least on A” unless specifically recited differently.

[0167] The following aspects are illustrative only and may be combined with other aspects or teachings described herein, without limitation.

[0168] Aspect 1 is a method of wireless communication at a first user equipment (UE), including: obtaining a configuration associated with a discontinuous reception (DRX) cycle of a second UE, where the configuration includes a first indication of a first set of resources and at least one condition, where the first set of resources is different from a second set of resources utilized for the DRX cycle of the second UE; and transmitting, to the second UE based on an occurrence of the at least one condition, sidelink (SL) data via the first set of resources or the second set of resources.

[0169] Aspect 2 is the method of aspect 1, where the first set of resources is associated with a DRX on-duration of the second UE, where the second set of resources is associated with an extension of the DRX on-duration.

[0170] Aspect 3 is the method of aspect 2, where the extension of the DRX on-duration is triggered via a second indication.

[0171] Aspect 4 is the method of aspect 3, where the second set of resources is activated based on the second indication.

[0172] Aspect 5 is the method of any of aspects 2-4, where a time period of the extension of the DRX on-duration is included in the configuration or the time period of the extension of the DRX on-duration is based on a difference between a first time period of a configured DRX on-duration and a second time period corresponding to an active time in the DRX on-duration.

[0173] Aspect 6 is the method of any of aspects 2-5, where the at least one condition includes: a lack of a detection of first stage sidelink control information (SCI-1) during the DRX on-duration by the second UE, a lack of a detection of second stage sidelink control information (SCI-2) during the DRX on-duration by the second UE, or a detection of the SCI-2 during the DRX on-duration by the second UE.

[0174] Aspect 7 is the method of aspect 6, where the SCI-2 includes a second indication of a source identifier (ID) that corresponds to the first UE.

[0175] Aspect 8 is the method of aspect 6, where the SCI-2 includes a second indication of a source identifier (ID) that corresponds to the first UE and a destination ID that corresponds to the second UE.

[0176] Aspect 9 is the method of any of aspects 6-7 or aspects 6 and 8, where the extension of the DRX on-duration is activated immediately upon the second UE detecting the SCI-2 during the DRX on-duration, or where the extension of the DRX on-duration is activated upon the second UE detecting the SCI-2 during the DRX on-duration and after the DRX on-duration has expired.

[0177] Aspect 10 is the method of aspect 1-9, further including: transmitting, to the second UE, the configuration associated with the DRX cycle prior to transmitting the SL data.

[0178] Aspect 11 is the method of any of aspects 1-10, where the SL data is SL control signaling.

[0179] Aspect 12 is the method of any of aspects 1-11, further including: transmitting, to the second UE, a second indication of the at least one condition via sidelink control information (SCI) or PC-5 radio resource control (RRC) signaling.

[0180] Aspect 13 is an apparatus for wireless communication at a first UE including a memory and at least one processor coupled to the memory and based at least in part on information stored in the memory, the at least one processor is configured to perform a method in accordance with any of aspects 1-12.

[0181] Aspect 14 is an apparatus for wireless communications, including means for performing a method in accordance with any of aspects 1-12.

[0182] Aspect 15 is the apparatus of aspect 13 or 14 further including at least one of a transceiver or an antenna coupled to the at least one processor, where the at least one processor is configured to transmit the SL data via at least one of the transceiver or the antenna.

[0183] Aspect 16 is a computer-readable medium (e.g., a non-transitory computer-readable medium) including instructions that, when executed by an apparatus, cause the apparatus to perform a method in accordance with any of aspects 1-12.

[0184] Aspect 17 is a method of wireless communication at a second user equipment (UE), including: monitoring for sidelink (SL) data via a first set of resources, where the first set of resources is different from a second set of resources utilized for a discontinuous reception (DRX) cycle of the second UE; and receiving, from a first UE based on an occurrence of at least one condition, the SL data via the first set of resources or the second set of resources.

[0185] Aspect 18 is the method of aspect 17, where the first set of resources is associated with a DRX on-duration of the second UE, where the second set of resources is associated with an extension of the DRX on-duration.

[0186] Aspect 19 is the method of aspect 18, where the extension of the DRX on-duration is triggered via a second indication.

[0187] Aspect 20 is the method of aspect 19, where the second set of resources is activated based on the second indication.

[0188] Aspect 21 is the method of any of aspects 18-20, where a time period of the extension of the DRX on-duration is included in a configuration associated with the DRX cycle of the second UE.

[0189] Aspect 22 is the method of any of aspects 18-21, further including: determining a time period of the extension of the DRX on-duration based on a difference between a first

time period of a configured DRX on-duration and a second time period corresponding to an active time in the DRX on-duration.

[0190] Aspect 23 is the method of any of aspects 18-22, further including: detecting the occurrence of the at least one condition; and monitoring for the SL data via the second set of resources based upon detecting the occurrence of the at least one condition.

[0191] Aspect 24 is the method of aspect 23, where the at least one condition includes: a lack of a detection of first stage sidelink control information (SCI-1) during the DRX on-duration by the second UE, a lack of a detection of second stage sidelink control information (SCI-2) during the DRX on-duration by the second UE, or a detection of the SCI-2 during the DRX on-duration by the second UE.

[0192] Aspect 25 is the method of aspect 24, where the SCI-2 includes a second indication of a source identifier (ID) that corresponds to the first UE.

[0193] Aspect 26 is the method of aspect 24, where the SCI-2 includes a second indication of a source identifier (ID) that corresponds to the first UE and a destination ID that corresponds to the second UE.

[0194] Aspect 27 is the method of any of aspects 24-25 or aspects 24 and 26, where the extension of the DRX on-duration is activated immediately upon the second UE detecting the SCI-2 during the DRX on-duration, or where the extension of the DRX on-duration is activated upon the second UE detecting the SCI-2 during the DRX on-duration and after the DRX on-duration has expired.

[0195] Aspect 28 is the method of any of aspects 17-27, further including: receiving, from the first UE, a configuration associated with the DRX cycle prior to receiving the SL data.

[0196] Aspect 29 is the method of any of aspects 17-28, where the SL data is SL control signaling.

[0197] Aspect 30 is the method of any of aspects 17-29, further including: receiving a second indication of the at least one condition via sidelink control information (SCI) or PC-5 radio resource control (RRC) signaling.

[0198] Aspect 31 is an apparatus for wireless communication at a second UE including a memory and at least one processor coupled to the memory and based at least in part on information stored in the memory, the at least one processor is configured to perform a method in accordance with any of aspects 17-30.

[0199] Aspect 32 is an apparatus for wireless communications, including means for performing a method in accordance with any of aspects 17-30.

[0200] Aspect 33 is the apparatus of aspect 31 or 32 further including at least one of a transceiver or an antenna coupled to the at least one processor, where the at least one processor is configured to receive the SL data via at least one of the transceiver or the antenna.

[0201] Aspect 34 is a computer-readable medium (e.g., a non-transitory computer-readable medium) including instructions that, when executed by an apparatus, cause the apparatus to perform a method in accordance with any of aspects 17-30.

1. An apparatus for wireless communication at a first user equipment (UE), comprising:

a memory; and

at least one processor coupled to the memory and, based at least in part on information stored in the memory, the at least one processor is configured to:

- obtain a configuration associated with a discontinuous reception (DRX) cycle of a second UE, wherein the configuration includes a first indication of a first set of resources and at least one condition, wherein the first set of resources is different from a second set of resources utilized for the DRX cycle of the second UE; and
- transmit, to the second UE based on an occurrence of the at least one condition, sidelink (SL) data via the first set of resources or the second set of resources.
2. The apparatus of claim 1, wherein the first set of resources is associated with a DRX on-duration of the second UE, wherein the second set of resources is associated with an extension of the DRX on-duration.
 3. The apparatus of claim 2, wherein the extension of the DRX on-duration is configured to be triggered via a second indication.
 4. The apparatus of claim 3, wherein the second set of resources is configured to be activated based on the second indication.
 5. The apparatus of claim 2, wherein a time period of the extension of the DRX on-duration is included in the configuration or the time period of the extension of the DRX on-duration is based on a difference between a first time period of a configured DRX on-duration and a second time period corresponding to an active time in the DRX on-duration.
 6. The apparatus of claim 2, wherein the at least one condition includes:
 - a lack of a detection of first stage sidelink control information (SCI-1) during the DRX on-duration by the second UE,
 - a lack of a detection of second stage sidelink control information (SCI-2) during the DRX on-duration by the second UE, or
 - a detection of the SCI-2 during the DRX on-duration by the second UE.
 7. The apparatus of claim 6, wherein the SCI-2 includes a second indication of a source identifier (ID) that corresponds to the first UE.
 8. The apparatus of claim 6, wherein the SCI-2 includes a second indication of a source identifier (ID) that corresponds to the first UE and a destination ID that corresponds to the second UE.
 9. The apparatus of claim 6, wherein the extension of the DRX on-duration is configured to be activated immediately upon the detection of the SCI-2 during the DRX on-duration, or wherein the extension of the DRX on-duration is configured to be activated upon the detection of the SCI-2 during the DRX on-duration and after the DRX on-duration has expired.
 10. The apparatus of claim 1, wherein the at least one processor is further configured to:
 - transmit, to the second UE, the configuration associated with the DRX cycle prior to transmit the SL data.
 11. (canceled)
 12. The apparatus of claim 1, wherein the at least one processor is further configured to:
 - transmit, to the second UE, a second indication of the at least one condition via sidelink control information (SCI) or PC-5 radio resource control (RRC) signaling.
 13. The apparatus of claim 1, further comprising: at least one of a transceiver or an antenna coupled to the at least one processor, wherein to transmit the SL data, the at least one

processor is configured to transmit the SL data via at least one of the transceiver or the antenna.

14. A method of wireless communication at a first user equipment (UE), comprising:

- obtaining a configuration associated with a discontinuous reception (DRX) cycle of a second UE, wherein the configuration includes a first indication of a first set of resources and at least one condition, wherein the first set of resources is different from a second set of resources utilized for the DRX cycle of the second UE; and

- transmitting, to the second UE based on an occurrence of the at least one condition, sidelink (SL) data via the first set of resources or the second set of resources.

15. An apparatus for wireless communication at a second user equipment (UE), comprising:

- a memory; and

- at least one processor coupled to the memory and, based at least in part on information stored in the memory, the at least one processor is configured to:

- monitor for sidelink (SL) data via a first set of resources, wherein the first set of resources is different from a second set of resources utilized for a discontinuous reception (DRX) cycle of the second UE; and

- receive, from a first UE based on an occurrence of at least one condition, the SL data via the first set of resources or the second set of resources.

16. The apparatus of claim 15, wherein the first set of resources is associated with a DRX on-duration of the second UE, wherein the second set of resources is associated with an extension of the DRX on-duration.

17. (canceled)

18. (canceled)

19. (canceled)

20. The apparatus of claim 16, wherein the at least one processor is further configured to:

- determine a time period of the extension of the DRX on-duration based on a difference between a first time period of a configured DRX on-duration and a second time period corresponding to an active time in the DRX on-duration.

21. The apparatus of claim 16, wherein the at least one processor is further configured to:

- detect the occurrence of the at least one condition; and
- monitor for the SL data via the second set of resources based on the detection of the occurrence of the at least one condition.

22. The apparatus of claim 21, wherein the at least one condition includes:

- a lack of a detection of first stage sidelink control information (SCI-1) during the DRX on-duration by the second UE,

- a lack of a detection of second stage sidelink control information (SCI-2) during the DRX on-duration by the second UE, or

- a detection of the SCI-2 during the DRX on-duration by the second UE.

23. (canceled)

24. The apparatus of claim 22, wherein the SCI-2 includes a second indication of a source identifier (ID) that corresponds to the first UE and a destination ID that corresponds to the second UE.

25. The apparatus of claim **22**, wherein the extension of the DRX on-duration is configured to be activated immediately upon the at least one processor being configured to detect the SCI-2 during the DRX on-duration, or wherein the extension of the DRX on-duration is configured to be activated upon the at least one processor being configured to detect the SCI-2 during the DRX on-duration and after the DRX on-duration has expired.

26. (canceled)

27. (canceled)

28. (canceled)

29. (canceled)

30. (canceled)

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